



Review

Ionic liquids in separations of azeotropic systems – A review

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ABSTRACT

Efforts to make existing separation methods more efficient and eco-friendly may get a boost from the use of a relatively new class of compounds known as ionic liquids (ILs). The separation of azeotropic mixtures has conventionally been one of the most challenging tasks in industrial processes due to the fact that their separation by simple distillation is basically impossible.

This paper provides a critical review of methods using ILs as azeotrope breakers. Three separation processes were addressed: liquid–liquid extraction, extractive distillation, and supported liquid membranes. We examine the azeotrope breaking potential of ILs and compare their performance to that of conventional solvents. A systematic analysis of the influence of the structure of ILs on their azeotrope breaking capacity contributes to the establishment of guidelines for selecting the most suitable ILs for the separation of specific azeotropic mixtures.

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List of symbols

T	temperature	$V_{\text{Solute}}^{\text{Mix}}$	volume of solute in the initial azeotropic mixture (LLE data)
P	pressure	E^{EP}	extraction efficiency from extraction process data
az	azeotrope	$w_{\text{Inert}}^{\text{Feed}}$	mass fraction of the inert in feed stream (extraction process data)
IL, ILs	ionic liquid, ionic liquids	$w_{\text{Inert}}^{\text{Raffinate}}$	mass fraction of the inert in raffinate stream (extraction process data)
LLE	(liquid + liquid) equilibria	$w_{\text{Inert}}^{\text{Equilibrium}}$	mass fraction of the inert in one equilibrium stage (theoretical data)
VLE	(vapor + liquid) equilibria	SLMs, SILMs	supported liquid membranes, supported ionic liquid membranes
α_{ij}^{VLE}	relative volatility (component i /component j) from (vapor + liquid) equilibria data	S^{MEM}	selectivity from supported liquid membranes data
y_i, y_j	mole fraction of the component i or j in the vapor phase	$\text{Perm}_{\text{Solute}}$	permeability of the solute in supported liquid membranes
x_i, x_j	mole fraction of the component i or j in the liquid phase (for VLE data the mole fraction is ionic liquid free basis)	$\text{Perm}_{\text{Inert}}$	permeability of the inert in supported liquid membranes
$S_{\text{IL}}^{\text{VLE}}$	selectivity from (vapor + liquid) equilibria data	α^{MEM}	separation factor from supported liquid membranes data
$\alpha_{ij}^{\text{with IL}}$	relative volatility from VLE data using as entrainer ionic liquid	$p_{\text{Solute}}^{\text{Permeate}}$	partial pressure of the solute in the permeate
$\alpha_{ij}^{\text{without IL}}$	relative volatility from VLE data without ionic liquid (azeotrope)	$p_{\text{Inert}}^{\text{Permeate}}$	partial pressure of the inert in the permeate
β^{LLE}	distribution coefficient from (liquid + liquid) equilibria data	$p_{\text{Solute}}^{\text{Feed}}$	partial pressure of the solute in the feed
$x_{\text{Solute}}^{\text{IL Phase}}$	mole fraction of the solute in ionic liquid phase (LLE data)	$p_{\text{Inert}}^{\text{Feed}}$	partial pressure of the inert in the feed
$x_{\text{Solute}}^{\text{Organic Phase}}$	mole fraction of the solute in organic phase (LLE data)	w, wt	mass fraction
S^{LLE}	selectivity from (liquid + liquid) equilibria data	THF	tetrahydrofuran
$x_{\text{Inert}}^{\text{IL Phase}}$	mole fraction of the inert in ionic liquid phase (LLE data)	t	ton
$x_{\text{Inert}}^{\text{Organic Phase}}$	mole fraction of the inert in organic phase (LLE data)	ϵ	euro
E^{LLE}	extraction efficiency from (liquid + liquid) equilibria data	h	hour
$C_{\text{Solute}}^{\text{IL Phase}}$	concentration of solute in ionic liquid phase (LLE data)	mol	mole
$V_{\text{Solute}}^{\text{IL Phase}}$	volume of solute in ionic liquid phase (LLE data)	n	alkyl chain length
$C_{\text{Solute}}^{\text{Mix}}$	concentration of solute in the initial azeotropic mixture (LLE data)	ETBE	ethyl tert-butyl ether
		C_6, C_7	hexane, heptane

1. Introduction

1.1. Azeotropes

Increasing concern about environmental issues, as well as the establishment of new regulations, has recently directed the attention of the scientific community to novel processes based on greener technologies. In many areas of industry, solvent mixtures accumulate due to recycling difficulties. The separation of these mixtures into their pure components is necessary so that they can be reused. However, most of these solvent mixtures contain azeotropes and, thus, their separation by simple distillation becomes impossible. Azeotropes are involved in common chemical processes including the production of methyl acetate, tetrahydrofu-

ran, ter-methyl amyl ether, isopropanol, and vinyl acetate. Additionally, due to their growing popularity, biofuel processes typically produce fermented products that form azeotropes with water which is abundant in the fermentator. Among those products, ethanol is the most important example, due to its excellent properties as alternative fuel. However, compounds such as butanol also form aqueous azeotropic mixtures.

Several potential processes for separating azeotropes have come to light such as azeotropic distillation, extractive distillation, pressure swing distillation, liquid–liquid extraction, adsorption, membranes. Extractive distillation is the most common process for removing one of the components in the azeotropic system. Its greatest constraint is the tremendous energy costs required to achieve a fluid phase system. Therefore, liquid–liquid separation,

TABLE 1

Azeotropes studied using ionic liquids as separating agent (azeotrope data from reference [1]).

Component 1	Component 2	Component 3	T_{az}/K	$y_{1,az}$	$y_{2,az}$	P_{az}/kPa	Type
<i>Aqueous systems</i>							
(Water + alcohols)							
Water	Ethanol	Ethyl acetate	351.3	0.103	0.084	101.33	OX
Water	Ethanol		343.4	0.09		101.33	OX
Water	2-Propanol		353.7	0.326		101.33	OX
Water	1-Propanol		360.8	0.568		101.33	OX
Water	2-Methyl-2-propanol		353.0	0.401		101.33	OX
Water	1-Butanol		365.5	0.754		101.33	EX
(Water + tetrahydrofuran)							
Water	Tetrahydrofuran		336.7	0.183		101.33	OX
<i>Alcoholic systems</i>							
(Alcohols + esters)							
Methanol	Methyl acetate		328.2	0.348		107.19	OX
Ethanol	Ethyl acetate		344.9	0.459		101.33	OX
1-Propanol	1-Propyl acetate		367.9	0.619		101.33	OX
2-Propanol	Ethyl acetate		348.0	0.305		101.33	OX
1-Butanol	1-Butyl acetate		390.0	0.770		101.33	OX
(Alcohols + ketones)							
Methanol	Acetone		328.3	0.240		101.33	OX
Ethanol	2-Butanone		347.2	0.508		101.33	OX
2-Propanol	2-Butanone		350.6	0.383		101.33	OX
(Alcohols + aliphatics hydrocarbons)							
Methanol	Heptane		332.0	0.728		101.33	OX
Ethanol	Hexane		331.7	0.341		101.33	OX
Ethanol	Heptane		345.2	0.668		101.33	EX
(Alcohols + halogenated)							
Methanol	Chloroform		328.2	0.352		107.99	OX
Ethanol	Chloroform		332.5	0.159		101.33	OX
<i>(Aromatic + aliphatic hydrocarbons) systems</i>							
Benzene	Hexane		341.5	0.050		101.33	OX
Benzene	Cyclohexane		353.2	0.546		109.18	OX
Benzene	Heptane		353.3	0.992		101.32	OX
Pyridine	Heptane		368.6	0.300		101.33	OX
<i>Other systems</i>							
Methanol	Dimethyl carbonate		337.3	0.850		102.52	OX
Ethanol	Ethyl tert-butyl ether		340.0	0.373		101.72	OX
Hexane	Ethyl acetate		338.0	0.657		101.32	OX
Cyclohexane	2-Butanone		345.0	0.562		101.33	OX

O: homogeneous azeotrope in a completely miscible system.

E: heterogeneous azeotrope.

X: pressure maximum.

based on the immiscibility between two liquid phases at room temperature, has emerged as a more economical and environmentally beneficial option.

The thermophysical properties of ionic liquids (ILs), as described below, enable them to make important contributions to azeotropic breaking. Table 1 presents a list of the systems in which ILs were used as separating agents together with the azeotropic point (T, P, y) coordinates [1].

A recent upsurge in publications published until 2010 regarding azeotrope breaking using ILs has created the need for this review article, which limits itself to considering direct methods. Despite their potential to provide valuable insights into the interactions between the organic compounds and ionic liquids, important sources of experimental data such as activity coefficients at infinite dilution and carbon dioxide solubility, as well as models (most commonly activity coefficients models, UNIQUAC, NRTL and COSMO-RS) plus molecular simulation studies are beyond the scope of this work.

1.2. Ionic liquids

Ionic liquids have been recognized for about a century, but have only started receiving closer attention in the last two decades. They have recently become one of the increasingly popular “green”

media for engineers [2,3], not only due to their remarkable physicochemical properties but also for their recyclability. Moreover, ILs can be tailored for specific applications by accurately selecting the chemical nature of both the cation and the anion [4]. This fact, combined with the immense number of possible combinations of cation and anion, have conferred upon them the designation of “designer solvents”. This feature is very attractive for chemical engineers since the fine-tuning of solvent properties allows them to optimize process efficiency and cost.

Ionic liquids were historically thought to be non-volatile, non-flammable thermally, and chemically stable salts. However, based on recent data, these assumptions have been progressively reconsidered [5,6]. The toxicity of ILs is also a hot issue that is driving the search for new environmentally friendly alternatives [7]. It is now commonly accepted that the label “green solvents” does not apply to all ILs in all situations. Nevertheless, a wide range of IL applications regularly provide either better yields or lower toxicities than conventional processes.

One of the relevant problems in the chemical industry is the separation of azeotropes. The remarkable azeotrope breaking capacity of ILs, their almost null volatility at room temperature [5], and their recycling ease, have granted them recognition as possible sustainable alternatives to conventional solvents. Moreover, the physical and corrosion properties of ILs are generally more

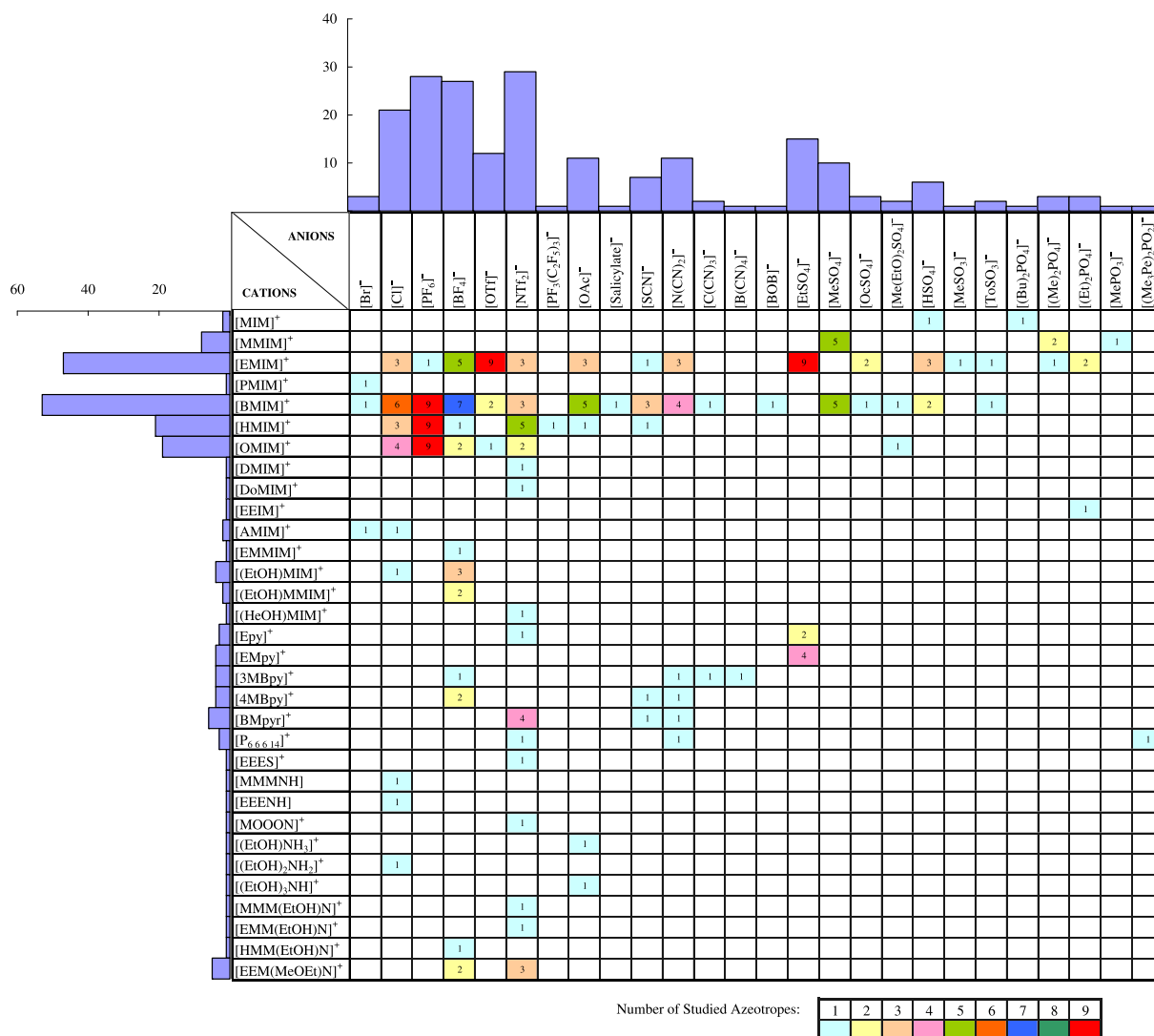


FIGURE 1. Mapping of azeotropes in the matrix of ILs anions and cations used as separating agents. The colors refer to the number of different studied azeotropes from literature per ion combination; the length of the bars gives the count per ion.

adequate for separation processes than those of inorganic molten salts, opening the door to their use in azeotropic separation processes.

Figure 1 maps the large number of publications already available on this subject showing the distinct azeotropes studied with different IL anions and cations used as separating agents. Table 2 lists the nomenclature used for all IL cations and anions mentioned herein. It is important to highlight the fact that most of the studies, approximately 75%, were performed using 1-alkyl-3-methylimidazolium based ILs. The range of anions utilized is generally much greater, with special emphasis on $[\text{Cl}]^-$, $[\text{PF}_6]^-$, $[\text{BF}_4]^-$, $[\text{OTf}]^-$, $[\text{NTf}_2]^-$ and $[\text{EtSO}_4]^-$.

The use of ILs as entrainers in separation technology is promising [8], and several patents [9] already protected its application as an entrainer in extractive or azeotropic distillation.

1.3. Energy and economic costs

Distillation is the most widely used separation process in the chemical industry. Typically, distillation columns and their support facilities account for about one-third of the total capital cost and more than half of energy consumption. Consequently, the design

and optimization of the distillation train have a critical impact on the economics of the entire process [10].

In extractive distillation, the entrainer does not need to be vaporized and can be separated easily in a secondary distillation column. However, in azeotropic distillation, both the entrainer and components must be vaporized. Moreover, large amounts of entrainer are usually needed, leading to large energy consumption as compared to extractive distillation, making the extractive option more attractive [11]. Other criteria, e.g., corrosion, prices, sources, etc. should also be taken into consideration and, from the strictly economic viewpoint, the use of the entrainer with the highest relative volatility will always translate into the lowest total annual cost (TAC) [12].

Conventional extractive distillation typically results in low levels of product purity since the entrainer coming from the bottom of the solvent recovery column contains impurities which influence the separation effect. Moreover, extractive distillation consumes much more energy than liquid–liquid extraction. It is thus advisable to combine extractive distillation with another separation process such as azeotropic distillation, liquid–liquid extraction and so on [13]. It is also important to mention that new hybrid alternatives are gaining advantage over traditional processes. As an example, economic analysis showed that for the dehydration

TABLE 2
Nomenclature of the ionic liquids used as separating agents.

Cations		Anions	
[MIM] ⁺	N-methylimidazolium	[Br] [−]	Bromide
[MMIM] ⁺	1-Methyl-3-methylimidazolium	[Cl] [−]	Chloride
[EMIM] ⁺	1-Ethyl-3-methylimidazolium	[PF ₆] [−]	Hexafluorophosphate
[PMIM] ⁺	1-Propyl-3-methylimidazolium	[BF ₄] [−]	Tetrafluoroborate
[BMIM] ⁺	1-Butyl-3-methylimidazolium	[OTf] [−]	Trifluoromethanesulfonate or Triflate or Trifluoromethylsulfonate
[HMIM] ⁺	1-Hexyl-3-methylimidazolium	[NTf ₂] [−]	Bis(trifluoromethylsulfonyl)imide
[OMIM] ⁺	1-Methyl-3-octylimidazolium	[PF ₃ (C ₂ F ₅) ₃] [−]	Tris(pentafluoroethyl)trifluorophosphate
[DMIM] ⁺	1-Decyl-3-methylimidazolium	[OAc] [−]	Acetate
[DoMIM] ⁺	1-Dodecyl-3-methylimidazolium	[Salicylate] [−]	Salicylate
[EEIM] ⁺	1-Ethyl-3-ethylimidazolium	[SCN] [−]	Thiocyanate
[AMIM] ⁺	1-Allyl-3-methylimidazolium	[N(CN) ₂] [−]	Dicyanamide
[EMMIM] ⁺	1-Ethyl-2,3-dimethylimidazolium	[C(CN) ₃] [−]	Tricyanomethanide
[(EtOH)MIM] ⁺	1-(2-Hydroxyethyl)-3-methylimidazolium	[B(CN) ₄] [−]	Tetracyanoborate
[(EtOH)MMIM] ⁺	1-(2-Hydroxyethyl)-2,3-dimethylimidazolium	[BOB] [−]	Bis[oxalato(2-)]-borate
[(HeOH)MIM] ⁺	1-(6-Hydroxyhexyl)-3-methylimidazolium	[EtSO ₄] [−]	Ethyl sulfate
[Epy] ⁺	1-Ethylpyridinium	[MeSO ₄] [−]	Methyl sulfate
[EMpy] ⁺	1-Ethyl-3-methylpyridinium	[OCSO ₄] [−]	Octyl sulfate
[3MBpy] ⁺	3-Methyl-N-butylpyridinium	[Me(EtO) ₂ SO ₄] [−]	Diethylenglycol monomethyl ether sulfate or 2-(2-methoxyethoxy)ethylsulfate
[4MBpy] ⁺	4-Methyl-N-butylpyridinium	[HSO ₄] [−]	Hydrogen sulfate
[BMpyr] ⁺	1-Butyl-1-methylpyrrolidinium	[MeSO ₃] [−]	Methanesulfonate
[P _{6 6 6 14}] ⁺	Trihexyl(tetradecyl)phosphonium	[ToSO ₃] [−]	p-Toluenesulfonate
[EEES] ⁺	Triethylsulfonium	[(Bu) ₂ PO ₄] [−]	Dibutylphosphate
[MOOON] ⁺	Methyltriethylammonium	[(Me) ₂ PO ₄] [−]	Dimethylphosphate
[(EtOH)NH ₃] ⁺	Monoethanolammonium	[(Et) ₂ PO ₄] [−]	Diethylphosphate
[(EtOH) ₂ NH ₂] ⁺	Diethanolammonium	[MePO ₃] [−]	Methylphosphonate
[(EtOH) ₃ NH] ⁺	Triethanolammonium	[(Me ₃ Pe) ₂ PO ₂] [−]	Phosphonium
[MMM(EtOH)N] ⁺	(2-Hydroxyethyl)trimethylammonium		Bis(2,4,4-trimethylpentyl)phosphinate
[EMM(EtOH)N] ⁺	Ethyl(2-hydroxyethyl)dimethylammonium		
[HMM(EtOH)N] ⁺	Hexyl(2-hydroxyethyl)dimethylammonium		
[EEM(MeOEt)N] ⁺	Diethylmethyl(2-methoxyethyl)ammonium		
<i>Chloroaluminate ionic liquids</i>			
[BMIM][Cl]-AlCl ₃	1-Butyl-3-methylimidazolium chloride–aluminum chloride		
[MMNH][Cl]-AlCl ₃	Trimethylamine hydrochloride–aluminum chloride		
[EENH][Cl]-AlCl ₃	Triethylamine hydrochloride–aluminum chloride		

of isopropanol, a hybrid process consisting of distillation followed by pervaporation with ceramic membrane, is the most economical process leading to a total cost reduction of approximately 49% [14].

The use of ILs as entrainers in extractive distillation can overcome the purity issues since, at the range of temperature used, the vapor pressure of the IL is still negligible. In extractive distillation, the heat demand in the reboiler can be reduced, for example, by lowering reflux ratios through the selection of favorable entrainers. This optimization promises savings of at least 25% as compared to conventional processes [8].

Extractive distillation with ILs as the separating agent can potentially result in high-purity products. ILs can overcome most of the problems associated with the use of solid salts such as dissolution, reuse, and transport. Other important advantages are: (1) no traces of entrainers in the distillate due to their nonvolatility under the required conditions; (2) significant reduction of heat duties due to high selectivities and easy regeneration of the entrainer; and (3) the possibility of tailoring the properties of the IL.

In respect to liquid–liquid separation, Meindersma and de Haan [15] showed that, for the aromatic/aliphatic separation, the investment (including extraction solvent) in a process using ILs is about 65% of that of the conventional sulfolane process and the heat load is only about 20%.

1.4. Separation process

The scope of the review is limited itself to discussing separation processes utilizing ILs as separating agents.

1.4.1. Extractive or azeotropic distillation

Extractive distillation is a common technique for removing one of the mixture components at or close to its azeotropic point. It involves the addition of a new heavy chemical compound (entrainer) which interacts with the components by altering their relative volatilities.

Organic solvents, inorganic salts or even hyperbranched polymers are the most commonly used entrainers. Extractive distillation with ILs as the separating agent integrates the advantages of a liquid solvent (easy operation) and a solid salt (high separation ability). Azeotropic distillation is also common in the separation of close boiling temperature or azeotropic mixtures; this technique requires the vaporization of both the entrainer and components into the top of an azeotropic distillation column. In this way, azeotropic distillation is not used when the entrainer is an ionic liquid [13].

Another advantage of ILs is their negligible vapor pressure at ambient temperature and pressure conditions. This minimizes the chance of loss to the atmosphere, thus reducing environmental and worker exposure issues; it further increases the ILs' recyclability and reusability, by stripping, evaporation or drying, thereby minimizing costs. The relative volatility, α_{ij}^{VLE} , and selectivity, S_{ij}^{VLE} , can be calculated from (vapor + liquid) equilibria (VLE) data and are defined as:

$$\alpha_{ij}^{VLE} = \frac{(y_i/x_i)}{(y_j/x_j)}, \quad (1)$$

$$S_{ij}^{VLE} = \frac{\alpha_{ij}^{with\ IL}}{\alpha_{ij}^{without\ IL}}. \quad (2)$$

1.4.2. Liquid–Liquid extraction

Liquid–liquid extraction is based on the immiscibility between two liquid phases and it is used to separate a dissolved component (solute) from its diluent (inert) by its transfer into a second component (extraction solvent) liquid phase. The equilibrium in extraction can be characterized by the values of distribution coefficient, β^{LLE} , and selectivity, S^{LLE} . These parameters are widely used in assessing the solvent feasibility in the liquid–liquid extraction and are defined as follows:

$$\beta^{\text{LLE}} = \frac{x_{\text{Solute}}^{\text{IL Phase}}}{x_{\text{Solute}}^{\text{Organic Phase}}} \quad (3)$$

$$S^{\text{LLE}} = \left(\frac{x_{\text{Inert}}^{\text{Organic Phase}}}{x_{\text{Inert}}^{\text{IL Phase}}} \right) \cdot \left(\frac{x_{\text{Solute}}^{\text{IL Phase}}}{x_{\text{Solute}}^{\text{Organic Phase}}} \right) \quad (4)$$

The distribution coefficient determines the amount of IL required for the extraction process and is thus related to the capacity of the IL while the selectivity evaluates the efficiency of the IL used as solvent. Selectivity also indicates the ease of extraction of a solute from the diluent (inert) and the distribution coefficient indicates the solute-carrying capacity of the solvent. Higher selectivity corresponds, theoretically, to fewer stages required for a given separation and less residual inert in the extract. A higher distribution coefficient corresponds to a lower solvent flowrate for given separation, a smaller-diameter column, and lower operating costs.

The extraction efficiency indicates the ability of the solvent, IL, to remove solute from the azeotropes. This parameter can either be calculated from (liquid + liquid) equilibria (LLE) data, E^{LLE} , or from extraction process data, E^{EP} , and is defined as follows:

$$E^{\text{LLE}} = \frac{C_{\text{Solute}}^{\text{IL Phase}} \cdot V_{\text{Solute}}^{\text{IL Phase}}}{C_{\text{Solute}}^{\text{Mix}} \cdot V_{\text{Solute}}^{\text{Mix}}} \cdot 100, \quad (5)$$

$$E^{\text{EP}} = \frac{w_{\text{Inert}}^{\text{Feed}} - w_{\text{Inert}}^{\text{Raffinate}}}{w_{\text{Inert}}^{\text{Feed}} - w_{\text{Inert}}^{\text{Equilibrium}}} \quad (6)$$

1.4.3. Supported liquid membranes

Membranes are functional materials for many separation processes including bioseparations. Such processes are energy efficient, can be operated under moderate temperature and pressure, and do not produce large amounts of waste solvents. Supported liquid membranes (SLMs) are porous solids impregnated with the liquid used for the separation. Conventional SLMs have two common problems, e.g., the loss of liquid through vaporization or its dissolution in the contacting phases and the loading limits [16,17]. Since ILs are nonvolatile and are able to selectively modify the properties of the membrane solvent, they have been advantageously incorporated into the porous membranes to form the so-called functional supported ionic liquid membranes (SILMs). SILMs are versatile in a number of applications, including azeotropic mixtures separations.

To clarify the controlling factor in the permeation process, the selectivity of the permeability, S^{MEM} , is calculated in the following equation:

$$S^{\text{MEM}} = \frac{\text{Perm}_{\text{Solute}}}{\text{Perm}_{\text{Inert}}} \quad (7)$$

The separation factor for the vapor permeation, α^{MEM} , is defined as follows:

$$\alpha^{\text{MEM}} = \frac{(p_{\text{Solute}}^{\text{Permeate}}/p_{\text{Inert}}^{\text{Permeate}})}{(p_{\text{Solute}}^{\text{Feed}}/p_{\text{Inert}}^{\text{Feed}})} \quad (8)$$

2. Azeotrope separation

2.1. Aqueous systems

2.1.1. (Water + ethanol) system

Due to its industrial importance and the availability of components, the (ethanol + water) system is by far one of the most studied. Ethyl alcohol is produced by two main processes, acid catalyzed hydration of ethylene and fermentation of sugars [18]. Hydration of ethylene is the primary method for the industrial production of ethyl alcohol, while fermentation produces most beverage alcohol. Industrially produced ethanol has many uses: fuel and gasoline additive [19]; as a solvent in the manufacture of varnishes, lakers, inks, paints, and cleaning products; in the preparation of essences and perfumes in the cosmetic industry [20]; as flavorings and thickeners in the food industry; as an extraction solvent and carrier in medicine and the pharmaceutical industry; and finally as a powerful disinfectant, anti-microbial and anti-staling compound [21]. All beverage alcohol and much of that used in industry are synthesized through fermentation of a variety of products including grains such as corn, potato mashes, fruit juices, and beet and cane sugar molasses [22]. It, therefore, is an important bio-refinery product. One of the major drawbacks of switching to a biomass-based industry is the production cost of the base alcohols. The high affinity of ethanol for water, which makes it particularly useful in industrial applications, turns out to be a problem in its purification. Purifying an alcohol requires about 6% of the energetic value of the compound itself, much of which is used in separating and concentrating the alcohol from the fermentation broth where it is produced [23].

Separating alcohols from water conventionally requires an energetically costly series of distillation columns. The (vapor + liquid) equilibrium of {ethanol + water} system has been subject of innumerable studies. In the Dortmund data bank there are 127 references to 245 data sets reported for this system [24]. At atmospheric pressure, ethanol + water demonstrates a minimum boiling temperature homogeneous azeotrope at $x_{\text{Ethanol}} = 0.897$ and $T = 351.25 \text{ K}$ (table 1). Many entrainers have been used. Among them, inorganic salts have been successfully implemented. Due to their polarity, inorganic salts can be easily dissolved in the mixture, creating a noticeable salt effect. In the database ELVM2003 there are 102 references to using 46 inorganic salts to move or break the ethanol + water azeotrope [25]. The most commonly used inorganic salts are calcium chloride and potassium acetate. Nevertheless, the use of salt distillation process may erode the sieve plate, precipitate and build up in the tower due to the causticity and limited solubility [26]. In order to avoid these problems conventional entrainers such as N-methylpyrrolidone (DMP), dimethylsulphoxide (DMSO) and sulpholane, are also used.

Table 3 lists research data (or calculated from them) regarding the ionic liquids used as azeotrope breakers for the (water + ethanol) system for extractive distillation, liquid–liquid extraction and supported liquid membrane [26–48].

2.1.1.1. Separation by extractive distillation. An analysis of the articles related to the use of entrainers for the (ethanol + water) binary system shows that most of the work has been done through the evaluation of VLE behavior of ternary systems of (ethanol + water + entrainer) so that the feasibility of extractive or azeotropic distillation technique can be proposed.

Most of the ILs studied so far as entrainers for the (ethanol + water) system show a salting out effect, which enhances the relative volatility of the ethanol and may even eliminate the azeotropic phenomena at a specific IL content.

TABLE 3

Ionic liquids used as azeotrope breakers on literature for the separating of water + ethanol, ethanol/water selectivity from LLE data (S^{LLE}), ethanol/water relative volatility from VLE data (α^{VLE}) at azeotropic composition and ethanol/water separation factor from membranes data (α^{MEM}).

Azeotrope	Ionic liquids	S^{LLE}	$\alpha^{\text{VLE } a}$	α^{MEM}	Remarks	Reference
Water + ethanol	[EMIM][OTf]		1.27		VLE data at 100 kPa	[27]
	[EMpy][EtSO ₄]		2.40 ^b		VLE data at 101.3 kPa	[28]
	[BMIM][Cl]		1.38		VLE data at 30, 40 100 kPa	[29]
	[BMIM][OTf]		1.71 ^c		VLE data at 314.2 and 331.7 K	[30]
	[OMIM][OTf]		0.95 ^d			
	[HMIM][Cl]		1.31		VLE data at 100 kPa	[31]
	[BMIM][MeSO ₄]		2.38 ^b		VLE data at 101.3 kPa	[32]
	[EMIM][EtSO ₄]		2.02 ^b		VLE data at 101.3 kPa	[33]
	[EMIM][BF ₄]		1.37		VLE data at 100 kPa	[34]
	[BMIM][BF ₄]		1.37			
	[EMIM][N(CN) ₂]		1.36			
	[BMIM][N(CN) ₂]		1.53			
	[EMIM][Cl]		1.68			
	[BMIM][Cl]		1.41			
	[EMIM][OAc]		1.63			
	[BMIM][OAc]		1.46			
	[HMIM][Cl]		0.39		VLE data at 101.3 kPa	[35]
	[BMIM][Cl]		1.87 ^b		VLE data at 101.3 kPa	[36]
	[MMIM][(Me) ₂ PO ₄]		1.17 ^e		VLE data at 101.3 kPa	[26]
	[EMIM][(Et) ₂ PO ₄]		1.41 ^e			
	[BMIM][Br]		1.06 ^e			
	[BMIM][Cl]		1.25 ^e			
	[BMIM][PF ₆]		1.19 ^e			
	[EMIM][BF ₄]		1.44		VLE data at 363.15 K	[37]
	[BMIM][BF ₄]		1.34			
	[BMIM][Cl]		1.55			
	[EEIM][(Et) ₂ PO ₄]		n.a.		Vapor pressure data	[38]
	[EMIM][EtSO ₄]		n.a.		Vapor pressure data	[39]
	[EMIM][(Me) ₂ PO ₄]		n.a.		Vapor pressure data	[40]
	[MMIM][(Me) ₂ PO ₄]		n.a.		Vapor pressure data	[41]
	[MMIM][(Me) ₂ PO ₄]		n.a.		Vapor pressure data	[42]
	[PMIM][Br]		n.a.		Vapor pressure data	[43]
	[(EtOH)NH ₃][OAc]		n.a.		Boiling temperature at 101.3 kPa	[44]
	[(EtOH) ₃ NH][OAc]		n.a.			
	[(EtOH) ₂ NH ₂][Cl]		n.a.			
	[P _{6 6 6 14}][N(CN) ₂]	n.a.			LLE data at 298.15 K	[45]
	[P _{6 6 6 14}][(Me) ₃ Pe) ₂ PO ₂]	n.a.				
	[HMIM][NTf ₂]		10 ^f		LLE data at 295 K	[46]
	[BMIM][PF ₆]		0.40 ^g		LLE data at 298.15 K	[47]
	[HMIM][PF ₆]		0.33 ^h			
	[OMIM][PF ₆]		0.29 ⁱ			
	[BMIM][PF ₆]			4–6	Supported liquid membranes: vapor permeation	[48]

^a Ethanol molar fraction ≈ 0.95 and IL molar fraction ≈ 0.10 .

^b Ethanol molar fraction ≈ 0.55 and IL molar fraction ≈ 0.10 .

^c Ethanol molar fraction ≈ 0.90 and IL molar fraction ≈ 0.50 .

^d Ethanol molar fraction ≈ 0.95 and IL molar fraction ≈ 0.80 .

^e Ethanol molar fraction ≈ 0.95 and IL molar fraction ≈ 0.05 .

^f Ethanol molar fraction in organic phase = 0.04.

^g Ethanol distribution coefficient at ethanol molar fraction = 0.97.

^h Ethanol distribution coefficient at ethanol molar fraction = 0.48.

ⁱ Ethanol distribution coefficient at ethanol molar fraction = 0.60.

2.1.1.1.1. Effect of the nature of the anion. The effect of the ionic liquid's anion nature on the relative volatility of the system at two ethanol concentrations was studied at $x_{\text{Ethanol}} = 0.1$ (supporting information figure S1) and 0.95 (figure 2). Imidazolium cation was chosen due to the wealth of available data. There is a large difference in the relative volatility at or near the azeotropic point and at low ethanol concentrations. The separation factors are much larger (around 10 for the most effective ILs) at low ethanol concentrations than at or near the critical point (around 2 for the most effective ILs). At the azeotropic point, the most promising IL anions are [Cl][−] and [OAc][−]. The dialkyl phosphate based ILs also perform very well at low concentrations of ethanol. Note that these anions are also present in the most effective inorganic salts used for this system [25].

It is well known that the anion has a stronger role in interaction with a given solvent than the cation. Thus, the knowledge and evaluation of the interactions between the anion and both components

in the mixture can be very valuable for understanding these low relative volatilities at the azeotropic point. However, the analysis of the activity coefficients of several ILs with water and with ethanol [34] shows that the interaction of the anions with water and with ethanol follow the same order: [OAc][−] $\approx$ [Cl][−] > [N(CN)₂][−] > [BF₄][−], canceling out most of the effect. This may explain the fact that no dramatic increase in the relative volatility has been observed.

2.1.1.1.2. Effect of the nature of the cation. Only two different cations, [EMIM]⁺ and [EMpy]⁺ with the anion [EtSO₄][−], have been tested as entrainers (see supporting information, figure S2) for the (ethanol + water) system [28,33]. At low ethanol concentrations, [EMIM][EtSO₄] leads to higher relative volatility than [EMpy][EtSO₄]; at $x_{\text{Ethanol}} = 0.2$, both ILs have similar effects on the relative volatility of this system.

2.1.1.1.3. Effect of the alkyl chain length. The length of the alkyl side chain of both the anion and the cation is another important feature.

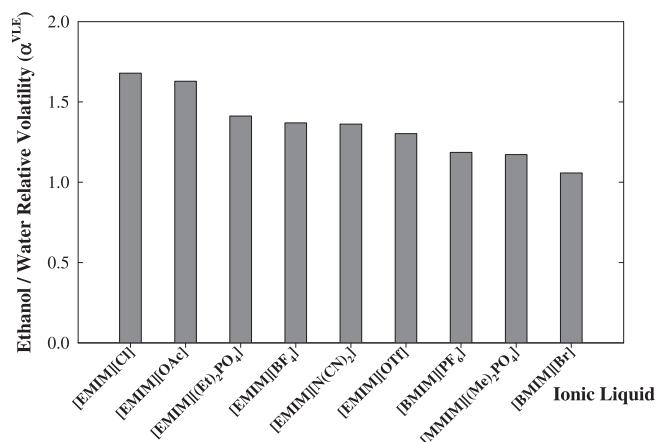


FIGURE 2. Effect of the anion nature in ILs based on imidazolium cation [26,27,32–34] taking into account the relative volatility for the (ethanol + water) system at $x_{\text{Ethanol}} \approx 0.95$ and $x_{\text{IL}} \approx 0.1$.

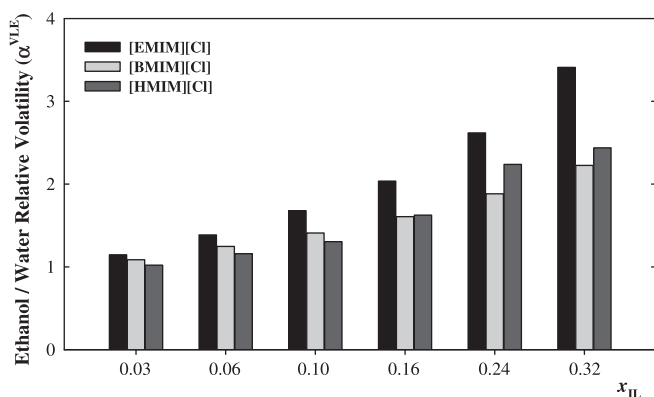


FIGURE 3. Effect of the IL cation chain length for chloride based ILs on the relative volatility for the (ethanol + water) system at $x_{\text{Ethanol}} = 0.95$. [EMIM][Cl] and [BMIM][Cl] data from [34]; [HMIM][Cl] from [31].

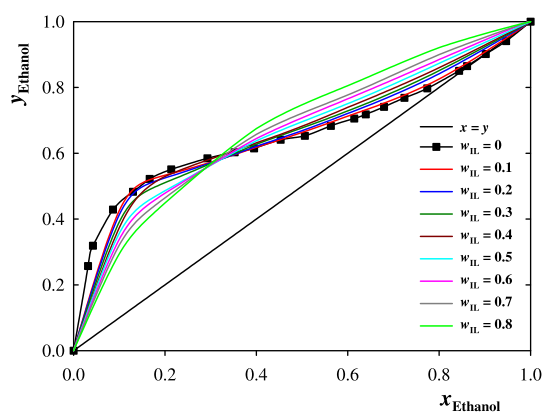


FIGURE 4. y versus x representation for (ethanol + water) system at variable concentrations of [HMIM][Cl] [31]. Ethanol + water data from [49].

The cation chain length effect ([EMIM]⁺ and [BMIM]⁺) was investigated for the following anions: [OAc][−], [N(CN)₂][−], [BF₄][−] [34]. For the [Cl][−] anion studies were performed using [EMIM]⁺ [34], [BMIM]⁺ [26,29,34,36,37], and [HMIM]⁺ [31,35]. Figure 3 represents studies conducted near the azeotropic point [31,34]. For all these ILs, the increase in the cation's chain length led to a less effective entrainer, since the relative volatility decreases at all

TABLE 4

Minimum molar fraction of IL needed to break the water + ethanol azeotrope at $x_{\text{Ethanol}} = 0.95$. All the data are from reference [34], except those for [EMIM][OTf] [27].

Cation	[Cl] [−]	[OAc] [−]	[N(CN) ₂] [−]	[BF ₄] [−]	[OTf] [−]
[EMIM] ⁺	0.015	0.013	0.019	0.018	0.023
[BMIM] ⁺	0.019	0.015	0.019	0.023	

concentrations. This indicates that small cations possess stronger IL + water interactions than large ones, forcing the ethanol into the vapor phase, thus increasing relative volatility. It is also worth noting that the lack of any significant effect of the IL cation's chain length at low IL concentrations ($x_{\text{IL}} < 0.03$).

An interesting study of the combined effect of the chain length in both the anion and the cation (see supporting information, figure S3), using [MMIM][(Me)₂PO₄] and [EMIM][(Et)₂PO₄] [26] leads to the same conclusions about the alkyl chain length. For $x_{\text{Ethanol}} > 0.40$, both ILs perform similarly, with relative volatilities approaching 1 at the azeotropic point; away from the azeotropic point ($x_{\text{Ethanol}} = 0.1$) and $x_{\text{IL}} < 0.2$, the smaller the IL the higher the relative volatility.

2.1.1.1.4. Effect of the ionic liquid concentration. Another important parameter to evaluate is the effect of IL concentration on the relative volatility of (ethanol + water) system. The following ILs were studied: [EMIM][OTf] ($x_{\text{IL}} = 0.05, 0.1$, and 0.2) [27]; [EMIM][BF₄] and [BMIM][BF₄] ($x_{\text{IL}} = 0.1, 0.3$, and 0.5) [37]; [MMIM][(Me)₂PO₄] and [EMIM][(Et)₂PO₄] ($w_{\text{IL}} = 0.1$ and 0.2) [26]; [BMIM][Cl] [29] and [HMIM][Cl] [31] ($x_{\text{IL}} = 0.1, 0.2, 0.4, 0.6, 0.8$, and 0.98). Behaviors differed according to the IL. Figure 4 clearly shows that for [HMIM][Cl] there is an inversion of behavior around $x_{\text{Ethanol}} = 0.4$; at low ethanol concentrations there is salting in effect while salting out occurs at high ethanol concentrations. The effect of [HMIM][Cl] [31] on the phase behavior (figure S4) is also interesting. While the boiling temperature of water is higher than that of ethanol, the boiling temperature of [HMIM][Cl] + water is lower than that of [HMIM][Cl] + ethanol at $w_{\text{IL}} = 0.8$. Although at $w_{\text{IL}} = 0.1$ the ethanol + water azeotrope is broken, it is clear that there is always a minimum for the bubble line and the dew line at a given w_{IL} .

In conclusion, the effect of the IL concentration on the relative volatility might not be as straightforward as expected since it is “slower” among different ILs. This is due to the balance of interactions between the IL–water and IL–ethanol [31].

Table 4 lists the minimum amount of IL needed to break the ethanol + water azeotrope for the several anions combined with [EMIM]⁺ and [BMIM]⁺ cations. Very small concentrations of each IL suffice. However, [EMIM][Cl] and [BMIM][Cl] have melting temperatures about (358 and 340) K [50], respectively, and high viscosities [51] and thus their use as an entrainers does not present the usual IL advantages over inorganic salts. Furthermore, [BMIM][OAc] starts to decompose at 446 K and has a viscosity of 440 mPa s at 298.15 K. The viscosity data for [EMIM][OAc] is above 129 mPa s. The solution viscosities of these four ILs with water and ethanol are close to the acceptable distillation limit of 100 mPa s, above which it becomes very difficult to manage the separation. [EMIM][OTf] has a melting temperature of 262.23 K [52], a viscosity of 41 mPa s [53] and a decomposition temperature higher than others, 613 K. Its only drawback is its high production cost.

2.1.1.1.5. Comparison with other entrainers. Figure 5 demonstrates an evaluation of the performance of several entrainers proposed for the (ethanol + water) system. The two polymers (hyperbranched poly(ethylene glycol) [54] and polyesteramide Hybrane S1200 [55]) and the inorganic salt (CaCl₂) [56] were chosen on the basis of the best separation factors for that particular family of entrainers. 1,2-Ethanediol [8] is the conventional entrainer and [BMIM][Cl] is one of the most promising novel candidates.

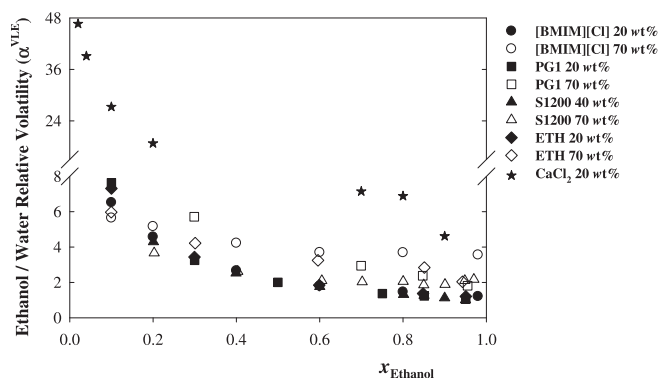


FIGURE 5. Comparison of the relative volatility for [BMIM][Cl] [29] and other proposed entrainers: hyperbranched polyglycerol (PG1) [54]; hybrane S1200 (S1200) [55]; calcium chloride (CaCl_2) [56]; 1,2-ethanediol (ETH) [8].

CaCl_2 is the most overall effective entrainer with higher separation factors than all the other compounds, especially at low ethanol concentrations. However, as mentioned before, when solid salt is used in industrial operations, dissolution, reuse and transport of salt present considerable challenges, not to mention the concurrent jam and erosion. Due to these issues, IL entrainers represent a step forward in the use of extractive distillation to break ethanol–water azeotrope. In addition, at the azeotropic point, both [BMIM][Cl] and CaCl_2 have very similar separation factors.

2.1.1.2. Separation by other methods. Other methods to identify and evaluate the suitability of ILs as entrainers were vapor pressure measurements (6 articles), boiling temperature (1 article), (liquid + liquid) equilibria (3 articles) and the use of supported ILs (1 article). Although vapor pressure and boiling temperature measurements are in fact VLE experiments, they were considered separately since the results lead to conclusions which differ from those of the VLE data.

As in the case of VLE experiments, the entrainer interacts with both solvents producing, occasionally, unexpected results. The addition of phosphate based 1,2 dialkylimidazolium ILs in the vapor pressure of the (ethanol + water) system was tested [41,42]. The obtained results are identical to those obtained when VLE measurements were performed [26]; the smaller the alkyl substituting hydrocarbon chain of the IL proves more effective. Binary data for ethanol and a series of phosphate-based ILs indicate that the ILs can be ordered as is $[\text{MMIM}][(\text{Me})_2\text{PO}_4] > [\text{EMIM}][(\text{Et})_2\text{PO}_4] > [\text{BMIM}][(\text{Bu})_2\text{PO}_4]$. Thus only the vapor pressure of ethanol + water + [MMIM][$(\text{Me})_2\text{PO}_4$] was measured. Figure 6 plots the tests of [EMIM][$(\text{Me})_2\text{PO}_4$] [40] and [EEIM][$(\text{Et})_2\text{PO}_4$] [38]. It can be observed that, close to the azeotropic point, both ILs perform equally well. The results from the first IL, [MMIM][$(\text{Me})_2\text{PO}_4$], [41,42] are similar to those obtained by VLE [26] with the salting-out of ethanol at all ethanol concentrations. However, for the second IL, salting-in occurs at low ethanol concentration while salting out is obtained near the azeotropic point. For [MMIM][$(\text{Me})_2\text{PO}_4$], this salting-in/salting-out change with the ethanol concentration also takes place. Vapor pressure data of [EMIM][EtSO_4] [39] and [PMIM][Br] [43] corroborate the trends for the VLE data.

Only three articles address the ethanol + water azeotrope using LLE. Chowdhury *et al.* [45] used long chain phosphonium based ILs. Chapeaux *et al.* [46] and Swatloski *et al.* [47] used [HMIM] $^+$ based ILs, with the $[\text{NTf}_2]^-$ and $[\text{PF}_6]^-$ anions respectively. For low concentrations of ethanol in water (typically $x_{\text{Ethanol}} = 0.04$ in a fermentation broth) the ethanol distribution coefficients are around 10. The authors conclude that large amounts of this IL are required to separate diluted ethanol aqueous solutions.

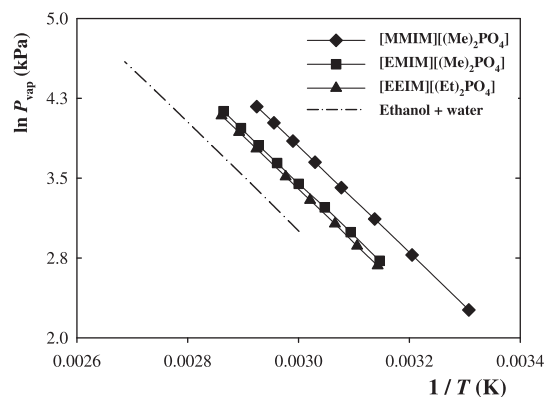


FIGURE 6. Experimental vapor pressure data for the systems (ethanol + water) (dashed-point line) [57] and (ethanol + water + IL) [41,38,40] at $x_{\text{Ethanol}} \approx 0.9$ (azeotropic point) and $x_{\text{IL}} \approx 0.05$.

Recent development in membrane technology has improved the efficiency of the (ethanol + water) system separation. [BMIM][PF_6] was supported on a porous membrane in a vapor permeation apparatus [48] with a resulting separation factor of 4 to 6 during 100 h. These results are very encouraging since the rapid fouling of the membranes is a major issue with this technique.

2.1.2. (Water + other alcohols) systems

This section deals with the azeotropic separation of mixtures of longer (alcohols + water). The alcohols already studied in the literature include 1-propanol, 2-propanol, 2-methyl-2-propanol and n-butanol. 1-Propanol is mainly used as a solvent, namely as a carrier and extraction solvent for natural products. Due to its solvent and antiseptic properties, it is also used in drugs and cosmetics [58]. 2-Propanol is also used in pharmaceutical, household, and personal products [59]. 2-Methyl-2-propanol is used for the removal of water from substances, in the extraction of drugs, in the manufacture of perfumes, in the recrystallization of chemicals, and as a chemical intermediate [60]. Butanol is an important bulk chemical with a wide range of industrial uses e.g., surface coatings, superabsorbents, fibers, plastics, resins, hard-surface cleaners, electronics, paints, as synthetic fruit flavoring product, pharmaceuticals and perfume manufacturing [61]. Mixtures of these longer alcohols with water occur frequently. Therefore, their recovery from aqueous solutions is a topic of great interest.

The study of azeotrope breaking using ILs in binary systems of longer alcohols and water is just beginning. Up to now, only 10 references of VLE [62–71], four of LLE [46,72–74], and one of membrane separation [48], have been found. In terms of ternary azeotropic mixtures, there is only one paper [75] dealing with (water + ethanol + ethyl acetate) system. Table 5 lists the ILs used as azeotrope breakers for the (water + alcohols) longer than ethanol binary azeotropes and for the (water + ethanol + ethyl acetate) ternary azeotrope [62–75].

2.1.2.1. Separation by extractive distillation. The results available highlight the feasibility of using ILs as entrainers for the separation of (water + alcohol) systems. As already shown for the (water + ethanol) system, most of the ILs showed a salting out effect, enhancing the relative volatility of the alcohol and even eliminating the azeotropic phenomena at specific IL contents.

2.1.2.1.1. Binary azeotropic mixtures. Effect of the nature of the anion and cation. The long list of commonly used anions with distinctive properties, requires the study of the anion effects on the chosen separation process. The study of azeotrope breaking of the system (water + 1-propanol) has only been carried out with

TABLE 5

Azeotropes in the literature regarding the separation of water + longer alcohols. ILs used as azeotrope breakers. 1-Butanol/water selectivity from LLE data (S^{LLE}). Alcohol/water relative volatility from VLE data (α^{VLE}) and 1-propanol/water separation factor from membranes data (α^{MEM}).

Azeotrope	Ionic liquids	S^{LLE} ^a	α^{VLE}	α^{MEM}	Remarks	Reference
Water + 1-propanol	[EMIM][OTf]		1.36 ^b		VLE data at 100 kPa	[62]
	[EMIM][BF ₄]		1.52 ^c		VLE data at 100 kPa	[63]
	[BMIM][BF ₄]		1.37 ^c			
	[BMIM][PF ₆]			6–8	Supported Liquid Membranes: Vapor Permeation	[48]
Water + 2-propanol	[EMIM][EtSO ₄]		1.41 ^d		VLE data at 333.15 K: headspace gas chromatography	[64]
	[BMIM][BF ₄]		1.51 ^d			
	[HMIM][NTf ₂]		1.41 ^e		VLE data at 332 K: headspace gas chromatography	[65]
	[BMPyr][NTf ₂]		1.51 ^e			
	[BMIM][BF ₄]		1.32 ^d		VLE data at 101.3 kPa	[66]
	[EMIM][BF ₄]		1.95 ^d		VLE data at 101.3 kPa	[67]
	[BMIM][BF ₄]		1.32 ^d		VLE data at 100 kPa	[68]
	[EMIM][BF ₄]		2.19 ^d		VLE data at 100 kPa	[63]
	[BMIM][OAc]		2.12 ^b		VLE data at 100 kPa	[69]
	[EMIM][OAc]		2.18 ^b			
	[BMIM][Cl]		2.20 ^b			
	[EMIM][N(CN) ₂]		1.74 ^b			
	[EMIM][BF ₄]		1.58 ^b			
	[BMIM][N(CN) ₂]		1.50 ^b			
	[BMIM][BF ₄]		1.38 ^b			
	[EMIM][NTf ₂]		n.a.		Vapor pressure data	[70]
	[BMIM][NTf ₂]		n.a.			
Water + 2-methyl-2-propanol	[EMIM][OAc]		3.28 ^b		VLE data at 100 kPa	[71]
	[BMIM][OAc]		2.58 ^b			
	[HMIM][OAc]		2.43 ^b			
	[EMIM][Cl]		3.58 ^b			
	[BMIM][Cl]		2.29 ^b			
	[HMIM][Cl]		2.05 ^b			
Water + 1-butanol	[(HeOH)MIM][NTf ₂]	40			LLE data at 295 K	[72]
	[HMIM][PF ₃ (C ₂ F ₅) ₂]	300				
	[HMIM][NTf ₂]	90			LLE data at 295 K	[46]
	[(EtOH)MIM][BF ₄]	n.a.			LLE data at 293 K	[73]
	[(EtOH)MIM][BF ₄]	n.a.				
	[BMIM][PF ₆]	21			LLE data at 296 K	[74]
	[OMIM][PF ₆]	37				
Water + ethanol + ethyl acetate	[BMIM][OAc]		14/84.1/11.3 ^f		VLE data at 100 kPa	[75]

^a 1-Butanol mass fraction in the initial aqueous solution ≈ 0.05 .

^b 2-Propanol molar fraction ≈ 0.95 and IL molar fraction ≈ 0.25 .

^c 1-Propanol molar fraction ≈ 0.43 and IL molar fraction ≈ 0.24 .

^d 2-Propanol molar fraction ≈ 0.67 and IL molar fraction ≈ 0.20 .

^e Water/2-propanol relative volatility; 2-propanol molar fraction ≈ 0.38 and IL molar fraction ≈ 0.95 .

^f Water molar fraction ≈ 0.04 and IL mass fraction ≈ 0.7 varying the (ethyl acetate molar fraction/(ethyl acetate molar fraction + ethanol molar fraction)) ratio. Results indicated are for relative volatility of ethyl acetate to ethanol/ethyl acetate to water/ethanol to water, respectively.

two distinct anions. The minimum molar fraction of IL needed to break the water + 1-propanol azeotrope at $x_{1\text{-Propanol}} \approx 0.43$ is smaller for [EMIM][BF₄], ≈ 0.30 [63], than for [EMIM][OTf], ≈ 0.34 [62]. The small relative volatility (see table 5) obtained for the [OTf][−] anion IL (compared to that of the [BF₄][−]) demonstrates its inferior entrainer capacity.

Results for the system (water + 2-propanol) cover a much larger spectrum of ILs. Three distinct experimental working conditions were selected for comparison, one at $x_{2\text{-Propanol}} \approx 0.95$, another close to the azeotrope composition and the last one at $x_{\text{IL}} = 0.95$. VLE data, namely with [EMIM][BF₄], [BMIM][BF₄], [EMIM][N(CN)₂], [BMIM][N(CN)₂], [EMIM][OAc], [BMIM][OAc], or [BMIM][Cl], were taken at $x_{2\text{-Propanol}} \approx 0.95$ [69] and are compared in figure 7. The results indicate that the effect on the relative volatility of 2-propanol to water is mainly dependent on the anion in the order: [Cl][−] > [OAc][−] > [N(CN)₂][−] > [BF₄][−]. This order is in agreement with that found for water + ethanol azeotrope.

Only Kim *et al.* [64], have performed studies with a distinct anion worth mentioning. Besides using [BMIM][BF₄], the authors have also used [EMIM][EtSO₄] as an azeotrope breaker for water + 2-propanol. At the concentrations used, the authors were not able to break the azeotrope, rather deviating it to higher concentrations of alcohol, from $x_{\text{Alcohol}} = 0.68$ to 0.88 and 0.84 using

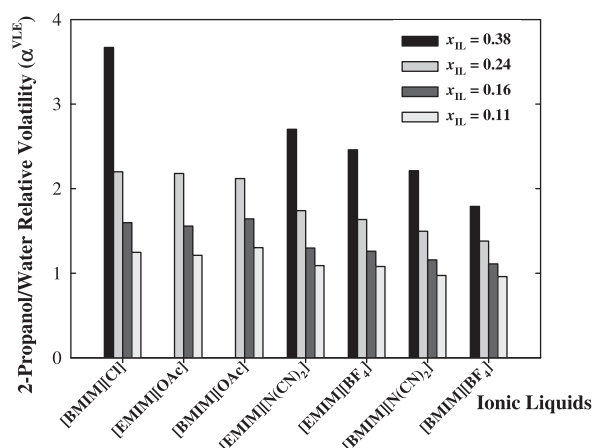


FIGURE 7. Effect of the IL [69] on the relative volatility for the (water + 2-propanol) system at $x_{2\text{-Propanol}} \approx 0.95$.

a mol fraction of 0.2 of [EMIM][EtSO₄] and [BMIM][BF₄], respectively. Direct comparison of the relative volatility (table 5) for

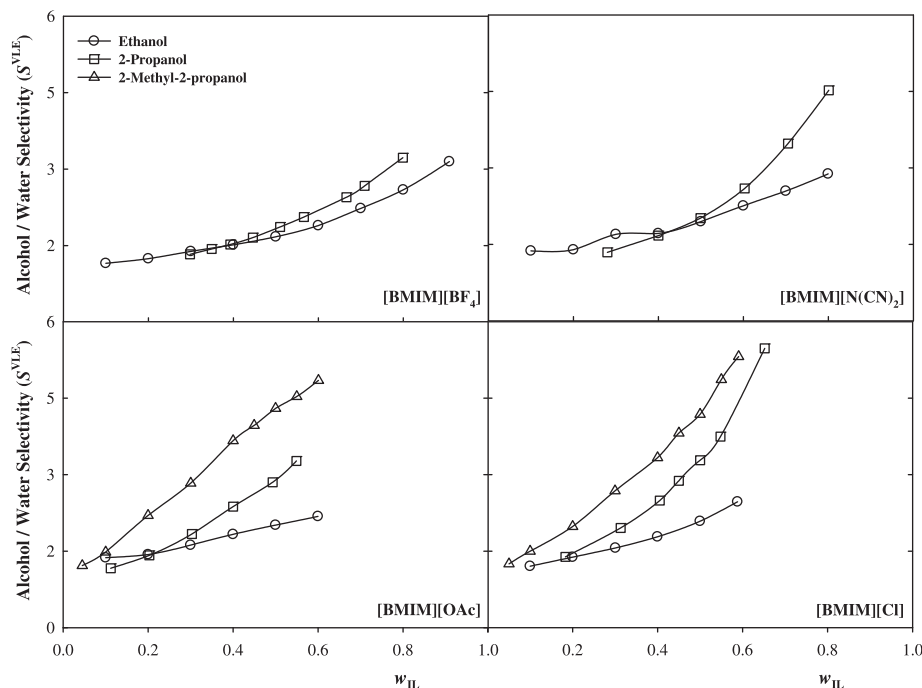


FIGURE 8. Effect of the alcohol size on the alcohol/water selectivity using distinct ILs [34,69,71] as azeotrope breaker at $x_{\text{Alcohol}} = 0.95$.

[BMIM][BF₄] and [EMIM][EtSO₄] shows that the [EtSO₄][−] anion is not a good option as an azeotrope breaker for this system.

The only report [71] about the azeotrope (water + 2-methyl-2-propanol) evaluates the effect of six distinct ILs (supporting information figure S5) based on the anions [Cl][−] and [OAc][−] and the cations [EMIM]⁺, [BMIM]⁺, [HMIM]⁺. The authors used a constant molar fraction of alcohol of 0.95 and the anion ordering is in close agreement with that for 2-propanol and ethanol.

The correct choice of the IL cation may also improve the separation process. However, regarding VLE data, Westerholt *et al.* [65] has been the only study to address the influence of the nature of the cation on the behavior of the azeotrope (water + 2-propanol) (supporting information figure S6). [HMIM][NTf₂] and [BMpyr][NTf₂] served as entrainers. It is important to stress the extremely high IL concentration used in this study, $x_{\text{IL}} \approx 0.85$ to 0.95. Since [NTf₂][−] based ILs are usually hydrophobic, the objective was to solubilize the alcohol and evaporate the water. The results clearly show that the proposed ILs, at the concentrations used, are not an alternative for this separation.

Effect of the alkyl chain length. Only one study of the impact of increasing the alkyl chain length of the cation in the azeotrope water + 1-propanol is available [63]. Two distinct 1-alkyl-3-methylimidazolium based cations with [BF₄][−] anion were used: [EMIM]⁺ and [BMIM]⁺. The results indicate that, at the same weight fraction of IL, $w_{\text{IL}} = 0.7$, the azeotrope is broken only in the case of the IL with [EMIM]⁺ cation. Thus, for this system, smaller alkyl chain length of the cation is more adequate for azeotrope breaking.

Figure 7 illustrates the effect of the alkyl chain length of the cation on the relative volatility of 2-propanol to water, namely the order [EMIM]⁺ > [BMIM]⁺. This is evident for all cation/anion pairs studied. The IL concentration effects on the relative volatility also appear. Several other articles studied [BF₄][−] based ILs with both [EMIM]⁺ and [BMIM]⁺ as cations as azeotrope breakers for the system (water + 2-propanol) [63,66–68]. All results confirm this trend, which is analogous to that found for the system (water + 1-propanol).

Comparison with other Entrainers. Data for the separation of propyl alcohols by extractive distillation using conventional

organic entrainers is available in patent literature [76,77]. Both patents study distinct alternatives and give examples of the best entrainers for extractive distillation. The experimental conditions in example 1 of Berg and Yang [76] (azeotrope 2-propanol + water with a molar fraction of the conventional entrainer (propyl butyrate) of approximately 0.15) are close to those shown in table 1 for 2-propanol. The relative volatility with propyl butyrate, 1.72, is lower than the average, 2.1, with [EMIM][BF₄] [63,67], which was not the best IL entrainer. In the case of the azeotrope 1-propanol + water, example 3 of Berg and Yang [76], the experimental conditions are also similar to those used with ILs as azeotrope breakers. In this case, the relative volatility with the conventional entrainer (butoxypropanol) is 2.2, which is higher than that obtained with [EMIM][BF₄]. This proves that ILs can perform better than conventional entrainers.

The addition of inorganic salts also breaks the azeotropes water + 1-propanol and water + 2-propanol by producing an important salting out effect over the alcohol [78–82]. In conclusion, inorganic salts are more effective than ILs, strictly from the azeotrope breaking point of view. Nevertheless, ionic liquids offer clear operational advantages.

Effect of the size of the alcohol. The experimental data highlights the effect of the alcohol hydrocarbon chain size in the separation of the azeotropes water + alcohols. To normalize the data, the relative volatility of each system was divided by the relative volatility of the binary system without entrainer at the same molar fraction obtaining the alcohol/water selectivity of each system. The binary systems data were obtained from the literature [49,62,63,83]. Figure 8 shows that the alcohol/water selectivities are close to unity at low IL mass fraction, but increase with increasing IL concentration. For all ILs, the following order is obtained: 2-methyl-2-propanol > 2-propanol > ethanol.

2.1.2.1.2. Ternary azeotropic mixture. Zhang *et al.* [75] analyzed the effect of [BMIM][OAc] on the ternary system (water + ethanol + ethyl acetate) at low water mole fraction. The experiments were performed at constant ethyl acetate/(ethanol + ethylacetate) ratio in the whole IL concentration range. As an example, at $w_{\text{IL}} = 0.7$ and $x_{\text{Water}} = 0.04$, the relative volatilities ethyl acetate to

TABLE 6

Ionic liquids used as azeotrope breakers in the literature for the separating of water + tetrahydrofuran, water/THF selectivity from LLE data (S^{LLE}) and THF/water relative volatility from VLE data (α^{VLE}).

Azeotrope	Ionic Liquids	S^{LLE} ^a	α^{VLE} ^b	Remarks	Reference
Water + Tetrahydrofuran	[EMIM][BF ₄]	15.58	4.40	VLE data and LLE data at 337.15 K: headspace gas chromatography	[37]
	[BMIM][BF ₄]	3.79	3.24		
	[OMIM][BF ₄]		2.50	VLE data at 337.15 K: headspace gas chromatography	[89]
	[BMIM][Cl]		37.26		
	[EMIM][Cl]		31.14		
	[EMIM][EtSO ₄]		13.59		
	[EMIM][ToSO ₃]		11.41		
	[EMIM][OcSO ₄]		12.87		
	[EMIM][OTf]		3.92		
	[EMIM][BF ₄]		4.35		
	[EMIM][PF ₆]		2.21		
	[EMIM][NTf ₂]		1.50		
	[BMIM][OAc]		36.86		
	[BMIM][Cl]		31.35		
	[BMIM][HSO ₄]		8.50		
	[BMIM][Salicylate]		8.33		
	[BMIM][SCN]		8.68		
	[BMIM][ToSO ₃]		9.45		
	[BMIM][N(CN) ₂]		8.33		
	[BMIM][MeSO ₄]		8.62		
	[BMIM][Me(EtO) ₂ SO ₄]		9.32		
	[BMIM][OcSO ₄]		22.03		
	[BMIM][BF ₄]		3.52		
	[BMIM][BOB]		15.67		
	[BMIM][NTf ₂]		1.26		
	[(EtOH)MIM][BF ₄]	14.79		LLE data at 293.15 K	[90]
	[(EtOH)MIM][Cl]	146.8			

^a Water molar fraction in organic phase ≈ 0.05 .

^b Azeotropic composition and IL molar fraction ≈ 0.30 .

ethanol, ethyl acetate to water and ethanol to water can reach 14, 84.1 and 11.3, respectively at distinct ethylacetate/(ethanol + ethylacetate) ratio. These results suggest the effectiveness of ILs as azeotrope breakers for ternary azeotropic systems.

2.1.2.2. Separation by other methods. As in the case of ethanol + water, vapor pressure measurements were considered separately since the information obtained does not allow for the same kind conclusions as VLE data. Doker and Gmehling [70] were the first to address the problem of azeotrope breaking of the water + 2-propanol at 353.15 K using different concentrations of [EMIM][NTf₂] and [BMIM][NTf₂] and these ternary systems show a positive deviation from Raoult's law. While the 2-propanol is miscible with the two ILs over the entire composition range, the two (IL + water) systems show an immiscibility gap beginning at a water mole fraction of approximately 0.5.

There is only one study [48] about separation of the azeotropes (water + alcohols) implementing supported liquid membranes. [BMIM][PF₆] was supported on the membranes and vapor permeation process was employed for the dehydration of aqueous 1-propanol. A separation factor of 6 to 8 during 270 h occurred. These results show a higher separation factor and experiment time than those mentioned previously for the water + ethanol. Supported liquid membranes seem to be a promising method for azeotropic separations.

The study of ILs as solvents for liquid–liquid extraction is scarce and devoted only to the (water + 1-butanol) system. Evaluating the distinct ILs that were tested for the azeotropic separation by LLE of 1-butanol and water (1-butanol/water selectivities in table 5), it is clear that the best option is the [HMIM][PF₃(C₂F₅)₃] which provides a higher distribution coefficient and selectivity.

2.1.3. (Water + tetrahydrofuran) system

Tetrahydrofuran (THF) dehydration is a process of special economical concern, due to increasing demand for anhydrous THF.

This is a cyclic ether with manifold applications, for example as a solvent in the manufacture of paint, adhesives, pharmaceutical, and other products, as an intermediate product and as a monomer [84]. Its most important industrial application includes the production of polytetramethylene-glycol, which is mandatory in the manufacture of polyurethane, polyester ether and Spandex fibers [85]. The THF market has grown approximately 5.6% annually [86] in recent years. It is produced by dehydrocyclization of 1,4-butanediol in the presence of an acid catalyst. The produced (THF + water) mixture presents an azeotrope at $x_{\text{THF}} = 0.817$ (table 1), and must be dehydrated to $x_{\text{THF}} = 0.999$ [87] for commercial use. The key step in purification is the breaking of the THF + water azeotrope [88], which is generally achieved either by pressure-swing (low/high pressure), extractive, or azeotropic distillation.

Table 6 [37,89,90] lists the ionic liquids used as azeotrope breakers for the (THF + water) system in the literature for extractive distillation and liquid–liquid extraction.

2.1.3.1. Separation by extractive distillation. Most of the research regarding the use of ILs as entrainers for the (THF + water) binary system studies the VLE behavior of ternary systems of (THF + water + IL) in order to evaluate the feasibility of extractive distillation techniques. Both the influence of systematic cation and the anion variation of imidazolium-based ILs with respect to their entrainer efficiency in extractive distillation have been investigated.

All ILs studied (from table 6) are suitable as entrainers for separating the azeotropic (THF + water) system by extractive distillation. Adding the selected IL to a binary (THF + water) system leads to a remarkable increase in the molar vapor fraction of THF, e.g., salting out effect, and thus the azeotrope is broken.

2.1.3.1.1. Effect of the alkyl chain length. The effect of the variation of the cation's alkyl chain length on the VLE of the azeotropic system (THF + water) is depicted in figure 9. The relative volatility of the low-boiling compound (THF) increases as the alkyl group is shortened from an octyl group over a butyl group to an ethyl

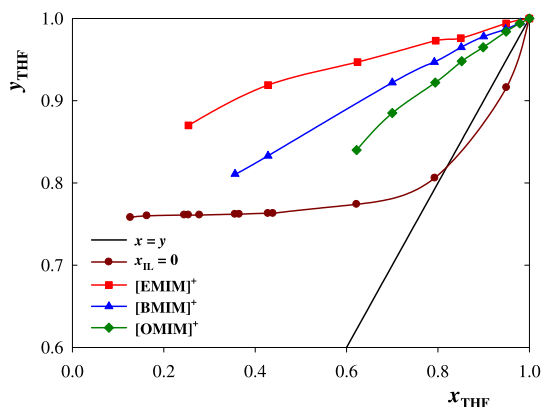


FIGURE 9. Influence of the variation of the length of the cation's alkyl chain on the VLE of the azeotropic system THF + water. Experimental VLE data of the systems (THF + water + $[\text{BF}_4]^-$) based ILs [37] at $x_{\text{IL}} = 0.5$ and 337.15 K; binary THF + water data [91] at 337.15 K.

group. If we accept that the IL entrainer suitability in extractive distillation rises with the increase in the relative volatility of the low-boiling component (with constant IL concentration) we can clearly see the entrainer superiority of $[\text{EMIM}][\text{BF}_4]$ as compared to corresponding imidazolium salts with longer alkyl chains.

Figure 10 depicts the effect of the IL concentration on the relative volatility of the (THF + water) system at azeotropic point ($x_{\text{THF}} = 0.817$). An increase in IL concentration leads to a higher THF concentration in the vapor phase and, therefore, to greater relative volatility due to strong selective interactions between the IL and water molecules. Figure 10 clearly shows that $[\text{EMIM}][\text{BF}_4]$ is the most suitable entrainer for the THF + water separation.

Although it is well known that $[\text{EMIM}][\text{BF}_4]$ is chemically unstable within aqueous media, Jork *et al.* [37] argued that this IL is a suitable entrainer for long-thermal applications in thermal separation. Despite its transition from slightly yellow to black, the ternary THF + water + $[\text{EMIM}][\text{BF}_4]$ VLE phase diagram was not found to change at all. This statement does not warrant the chemical stability of the IL investigated in a long-term application within aqueous media. Nevertheless, it was suggested that chemical reactions occurring in this mixture present minor impact on the observed thermodynamic behavior.

2.1.3.1.2. Effect of the nature of the anion. Effective extractive distillation entrainers must interact differently with the components of the system to be separated. Thus, the study of the influence of

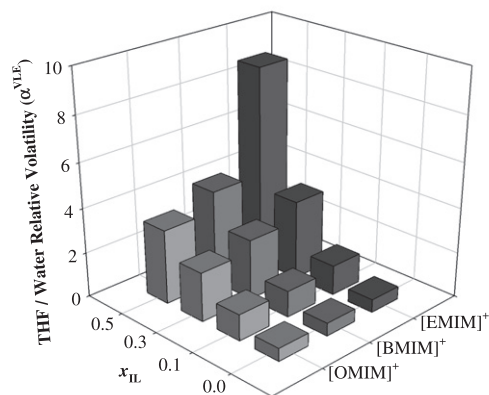


FIGURE 10. Influence of the IL cation chain length for $[\text{BF}_4]^-$ based ILs [37] on the relative volatility of the azeotropic system (THF + water) at $x_{\text{THF}} = 0.817$ (azeotropic point). Experimental binary THF + water data from [91].

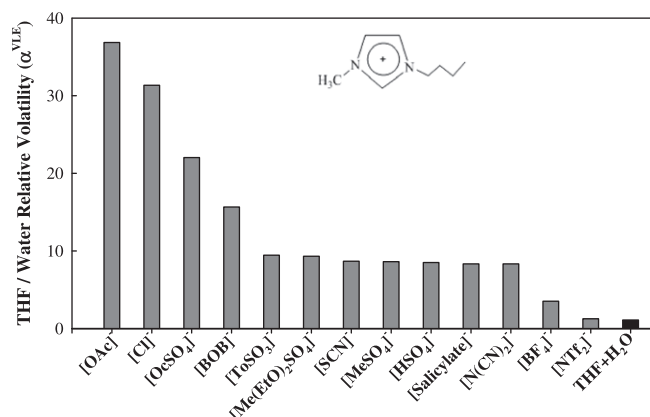


FIGURE 11. Effect of the IL anion's nature on the relative volatility of the (THF + water) azeotropic system at $x_{\text{THF}} = 0.817$ (azeotropic point) at 337.15 K [89] and $x_{\text{IL}} = 0.30$; binary THF + water data at 337.15 K [91]. The ILs cation is $[\text{BMIM}]^+$.

systematic anion variation of specific cation-based ILs with respect to their entrainer efficiency is an important first step towards the task-specific tailoring of ILs for extractive distillation.

The effect of the ionic liquid's anion nature on the relative volatility of the (THF + water) system at the azeotropic point ($x_{\text{THF}} = 0.817$) is plotted in figure 11 using the $[\text{BMIM}]^+$ cation (figure S7 illustrates the results for the $[\text{EMIM}]^+$ cation). The relative volatility for the binary (THF + water) azeotropic system also appears. All ILs in figure 11 are suitable entrainers for extractive distillation of azeotropic (THF + water) systems, since the addition of each IL to a binary (THF + water) system leads to a increase in the molar vapor fraction of THF. $[\text{BMIM}][\text{OAc}]$, $[\text{BMIM}][\text{Cl}]$, and $[\text{EMIM}][\text{OAc}]$ exhibit the highest entrainer efficiencies.

The relative volatilities of the THF + water + $[\text{BMIM}][\text{OAc}]$, $[\text{BMIM}][\text{Cl}]$, and $[\text{EMIM}][\text{Cl}]$ (figure 11 and supporting information figure S7) as compared to those of THF + water + $[\text{BMIM}][\text{BF}_4]$, $[\text{EMIM}][\text{BF}_4]$, and $[\text{OMIM}][\text{BF}_4]$ (figure 10) obviously demonstrate that the exchange of the anion results in a much larger change in the relative volatility of the low-boiling component than the modification of the cation's alkyl chain length. $[\text{BMIM}][\text{OAc}]$ is shown to be the most suitable for the THF + water separation. Moreover, the imidazolium salts containing the acetate ($[\text{OAc}]^-$) and chloride

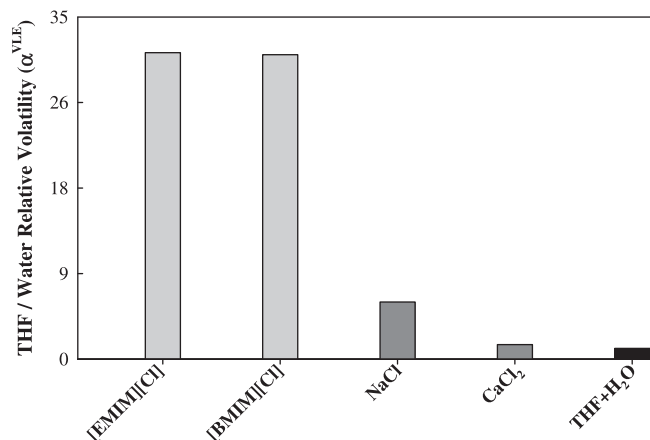


FIGURE 12. Effect of the chloride salt nature, inorganic versus IL, on the relative volatility of the (THF + water) azeotropic system at $x_{\text{THF}} = 0.817$ (azeotropic point). The relative volatilities were determined experimentally at 337.15 K. VLE data of the systems (THF + water + inorganic salts) saturated with each inorganic salt [91]; IL was always used at $x_{\text{IL}} = 0.30$ [89].

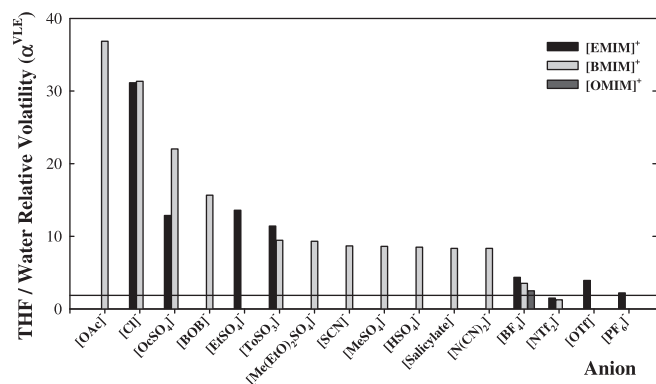


FIGURE 13. Effect of ILs and hyperbranched polymers on the relative volatility of the (THF + water) azeotropic system at $x_{\text{THF}} = 0.817$ (azeotropic point). The relative volatilities with ILs (mole fraction of 0.30) were determined experimentally at 337.15 K [89]. The straight horizontal line represents the relative volatility with hyperbranched polyesteramide Hydrane S1200 [92] (mole fraction of 0.40), determined at 343.15 K.

([Cl][−]) anions are the best of the studied anions regarding the selective interactions and the increasing relative volatility of the THF. This behavior is in accordance with the highest nucleophilicity (e.g., ability to hydrogen bond) of the [OAc][−] and [Cl][−] anions. The imidazolium salts containing the [OAc][−] and [Cl][−] anions selectively interact with water. The formation of hydrogen bonds between these ILs and water molecules reduces water activity and thus increases relative THF volatility.

2.1.3.1.3. Comparison with other entrainers. In order to fully address the entrainer efficiency of ILs in breaking the THF + water azeotrope, the performance of other entrainers such as inorganic chloride salts [91] and non-volatile hyperbranched polymers including hyperbranched polyesteramide Hydrane S1200 [92] must be compared.

The pioneering work of Sada *et al.* [91], containing VLE data, analyzes the effect of adding inorganic chlorides salts on the (vapor + liquid) equilibria of the (THF + water) azeotropic system at fixed solvent composition and atmospheric pressure. Figure 12 compares chloride inorganic salt versus IL regarding their effects on the relative volatility of the (THF + water) system at azeotropic point ($x_{\text{THF}} = 0.817$). The relative volatility for THF in the binary (THF + water) azeotropic system is also shown. It can be concluded that the chloride ILs ([EMIM][Cl] and [BMIM][Cl]) perform better than the chloride inorganic salts (NaCl and CaCl₂) in the salting-out of THF from the aqueous solution (supporting information figure S8), e.g., on increasing the relative volatility of the THF.

Some authors have recently suggested the use of nonvolatile hyperbranched polymers as entrainers for extractive distillation [54,55,93,94]. Therefore, a comparison between the influence of hyperbranched polymers and ILs on the VLE of (THF + water) azeotropic system is required to access which additive performs better in breaking the azeotrope. Figure 13 compares the effect of all investigated ILs, at a constant mole fraction of 0.30, on the relative volatility of THF in the (THF + water) system at azeotropic point ($x_{\text{THF}} = 0.817$) to that obtained for hyperbranched polyesteramide Hydrane S1200. The relative volatility for THF in the system (THF + water + hyperbranched polyesteramide Hydrane S1200) (mole fraction of 0.40) is 1.87 [92], and is represented as a straight horizontal line. It can be concluded that the hyperbranched polymer only performed better than two ILs, [EMIM][NTf₂] and [BMIM][NTf₂].

2.1.3.2. Separation by liquid–liquid extraction. An analysis of the literature regarding the use of ILs to break the THF + water azeotrope

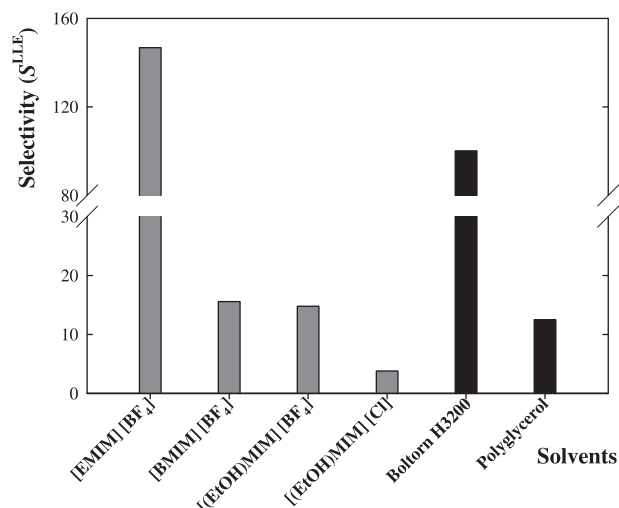


FIGURE 14. Selectivity versus extraction solvents used in the separation of the water + tetrahydrofuran azeotrope at solute molar fraction in organic phase ≈ 0.05 (for the ILs [37,90] solute = water and for the polymers [54,55] solute = tetrahydrofuran).

leads to different possible process options. The potential of ILs as extractive distillation entrainers for the (THF + water) azeotropic system has already been demonstrated. The aptness of ILs as extraction solvents for the (THF + water) separation will now be addressed.

Hu *et al.* studied the use of ILs as extraction solvent for the azeotrope water + THF [90]. Jork *et al.* [37] used ILs in the extractive distillation and solvent extraction for the separation of azeotrope. Figure 14 compares water/THF selectivity values for all the ILs tested as solvents in liquid–liquid extraction. The use of [(EtOH)MIM][Cl] as solvent leads to the highest selectivity values. The same figure compares ILs and polymers [54,55] used as extraction solvents. [(EtOH)MIM][Cl] continues to display the best results, although the hyperbranched polyester Boltorn H3200 presents better selectivity than other ILs.

The results, interestingly, also suggest the feasibility of specific tailoring of the extraction solvent. Figure 15 plots the contrast between IL and polymer behavior. While ILs are miscible in the water and immiscible in THF, the polymers are miscible in THF and immiscible in water. When the desired product is the THF, ILs may be used as solvents extracting water from azeotrope and obtaining THF in the upper phase. Otherwise, polymers are the best option.

2.2. Alcoholic systems

2.2.1. (Alcohols + ester) systems

In many chemical processes, (alcohol + ester) systems must be separated to obtain pure alcohol and pure ester. For example, these compounds constitute the products/reactants of further esterification reactions (esterification of methanol, ethanol, 1-propanol, and 1-butanol with acetic acid to obtain methyl acetate, ethyl acetate, 1-propyl acetate, and 1-butyl acetate). Azeotrope 2-propanol + ethyl acetate, for example, is important for the solvent extraction of edible oils [95]. The purification techniques of esters such as azeotropic distillation [96], extractive distillation [97,98], and membrane separation [99] have been investigated. Table 7 (data from [100–108]) reports on ILs studied as azeotrope breakers for separating (alcohols + ester) systems.

Ochilles *et al.* [100,103] studied extractive distillation for the separation of azeotropes methanol + methyl acetate and

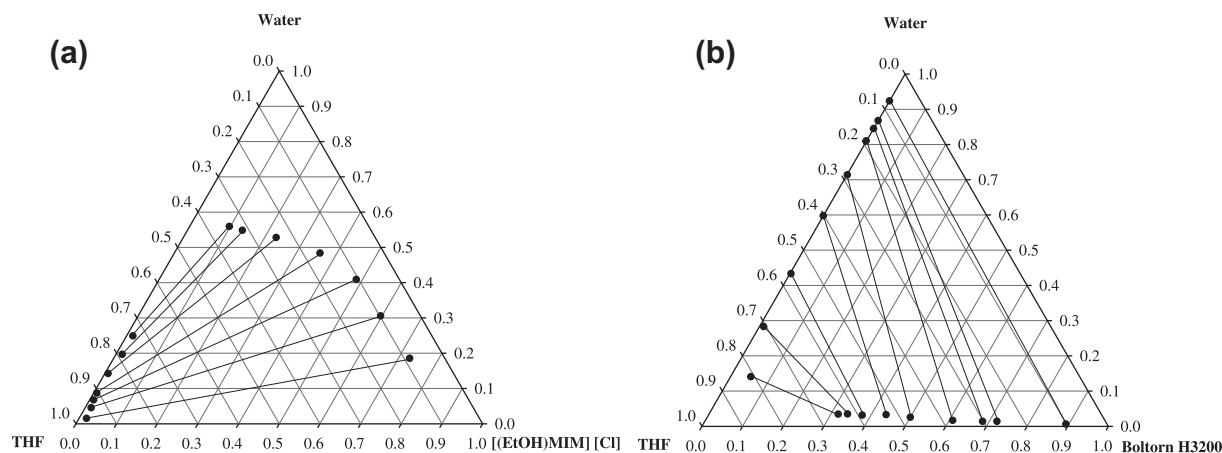


FIGURE 15. Experimental tie-lines in mass fraction of the ternary systems: (a) (tetrahydrofuran + water + [(EtOH)MIM][Cl]) at 293.15 K [90] and (b) (tetrahydrofuran + water + Hyperbranched polyester Boltorn H3200) at 334.15 K [55].

TABLE 7

Azeotropes in the literature regarding the separating of alcohols + esters. ILs used as azeotrope breakers. Alcohol/ester selectivity from LLE data (S^{LLE}) and acetate/alcohol relative volatility from VLE data (α^{VLE}).

Azeotrope	Ionic liquids	S^{LLE} ^a	α^{VLE} ^b	Remarks	Reference
Methanol + methyl acetate	[EMIM][OTf]		1.85	VLE data at 100 kPa	[100]
Ethanol + ethyl acetate	[EMIM][BF ₄]		1.83	VLE data at 101.32 kPa	[101]
	[BMIM][BF ₄]		1.45	VLE data at 101.32 kPa	[102]
	[OMIM][BF ₄]		2.35		
	[EMIM][OTf]		1.57	VLE data at 100 kPa	[103]
	[BMIM][Cl]	89 ^c		LLE data at 298.15 K	[104]
	[HMIM][Cl]	75 ^c			
	[OMIM][Cl]	97 ^c			
	[AMIM][Cl]	84 ^c			
	[AMIM][Br]	81 ^c			
	[EMIM][BF ₄]	1.56		LLE data at 298.15 K	[105]
	[(EtOH)MIM][BF ₄]	2.51			
	[EMMIM][BF ₄]	1.54			
	[(EtOH)MMIM][BF ₄]	1.93			
	[MIM][HSO ₄]	30.75		LLE data at 313.15 K	[106]
	[EMIM][HSO ₄]	19.61			
	[BMIM][HSO ₄]	10.26			
1-Propanol + 1-propyl acetate	[EMIM][HSO ₄]	21.51		LLE data at 313.15 K	[107]
2-Propanol + ethyl acetate	[BMIM][PF ₆]	3.29		LLE data at 298.15 K	[108]
	[HMIM][PF ₆]	2.65			
	[MMIM][MeSO ₄]	6.70			
1-Butanol + 1-butyl acetate	[EMIM][HSO ₄]	17.25		LLE data at 313.15 K	[107]

^a Alcohol molar fraction in organic phase ≈ 0.10 .

^b Azeotropic composition and IL molar fraction ≈ 0.30 .

^c Extraction efficiencies from equation (5).

ethanol + ethyl acetate using [EMIM][OTf] as entrainer. This IL showed a notable salting-out effect, enhancing the relative volatility of the ester to the alcohol and eliminating the azeotrope at a specific IL mole fraction: 0.129 and 0.20 for methanol + methyl acetate and ethanol + ethyl acetate, respectively.

Azeotrope ethanol + ethyl acetate is the most frequently studied system of this type. Figure 16 shows the influence of ILs on the relative volatility of ethyl acetate to ethanol. The separation ability of the four ILs at $x_{\text{IL}} = 0.30$ is in the following order: [OMIM][BF₄] > [EMIM][BF₄] > [EMIM][OTf] > [BMIM][BF₄] and is mainly due to the difference in polarity of the ILs and the demixing effect [102]. These results show that the tetrafluoroborates-based ILs are promising additives for the separation of ethylacetate + ethanol by extractive distillation.

However, the liquid–liquid extraction of (alcohol + ester) systems using ILs as solvents has been the most commonly used separation process, which may offer a high selectivity process with

low-energy consumption. An analysis of the alcohol/ester selectivities in table 7 suggests that the imidazolium-based ILs with hydrogen sulfate anion are the best extraction solvents for the separation of these azeotropes.

Ethanol + ethyl acetate azeotrope was also the most studied system for liquid–liquid extraction. Figure 17 illustrates the influence on the selectivity values of alkyl or hydroxyl substituent in the imidazolium cation of the ILs. These data show that [(EtOH)MIM][BF₄] has greater extraction capacity than the other ILs and that a hydroxyl group on the cation and a methyl group at the 2-position can obviously affect the IL ethanol interaction. The selectivity of the four ILs are ordered as follows: [(EtOH)MIM][BF₄] > [(EtOH)MMIM][BF₄] > [EMIM][BF₄] > [EMMIM][BF₄]. The selectivity of [EMIM][BF₄] is larger than [EMMIM][BF₄], and that of [(EtOH)MIM][BF₄] is clearly larger than [(EtOH)MMIM][BF₄]. [(EtOH)MMIM][BF₄] and [EMMIM][BF₄] have a methyl instead of hydrogen in the 2-position of the imidazolium cation greatly

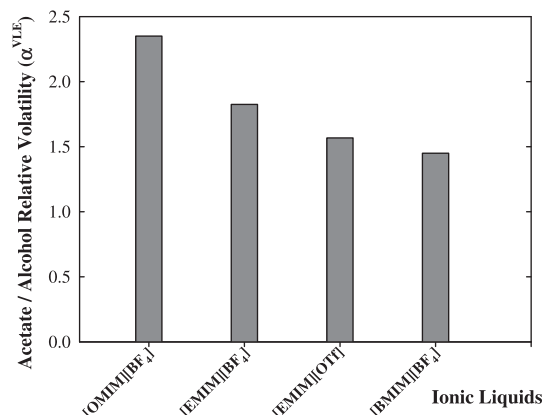


FIGURE 16. Acetate/alcohol relative volatility versus IL [101–103] used for (ethanol + ethyl acetate) system at azeotropic composition ($x_{\text{ethyl acetate}} \approx 0.54$) and $x_{\text{IL}} \approx 0.30$.

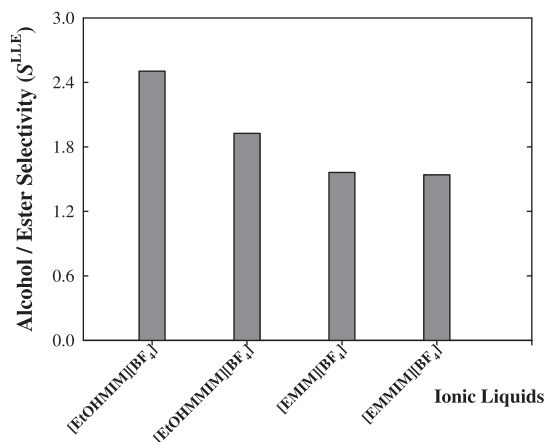


FIGURE 17. Alcohol/ester selectivity values versus ILs with the [BF₄][−] anion [105] used in the separation of the (ethanol + ethyl acetate) system at 298.15 K and ethanol molar fraction in organic phase ≈ 0.10 .

decreasing the ability of the cation to form a hydrogen bond with ethanol [105]. The selectivity of [(EtOH)MIM][BF₄] is clearly larger than that of [EMIM][BF₄], while that of [(EtOH)MMIM][BF₄] is obviously greater than [EMIM][BF₄]. The difference between the selectivity values results from the hydroxyl group in [(EtOH)MIM][BF₄] and [(EtOH)MMIM][BF₄], a fact which clearly enhances the interaction between the IL and ethanol (probably by hydrogen bonding).

Pereiro *et al.* [108] studied [MMIM][MeSO₄], [BMIM][PF₆] and [HMIM][PF₆] as extraction solvents in the separation of 2-propanol + ethyl acetate azeotrope. The binodal curve for the ternary system (2-propanol + ethyl acetate + [OMIM][PF₆]) was also determined. Figure 18 illustrates how the immiscibility region decreases with the increase in the length of the alkyl chain in the imidazolium ring when the [C_nMIM][PF₆] is used as solvent in liquid–liquid extraction. The analysis of the LLE data [108] indicates that the alkyl chain length of the imidazolium ring plays a negative role in the extraction solvent capability of [C_nMIM][PF₆], due, perhaps, to the hydrophobic steric effect of the alkyl group reducing the polar character of 2-propanol. However, the use of [MMIM][MeSO₄] as solvent leads to higher values of selectivity than those obtained using ILs involving [PF₆][−] as the anion, owing to the fact that the [MMIM][MeSO₄] IL contains just two methyl groups and the steric alkyl effect is, therefore, reduced.

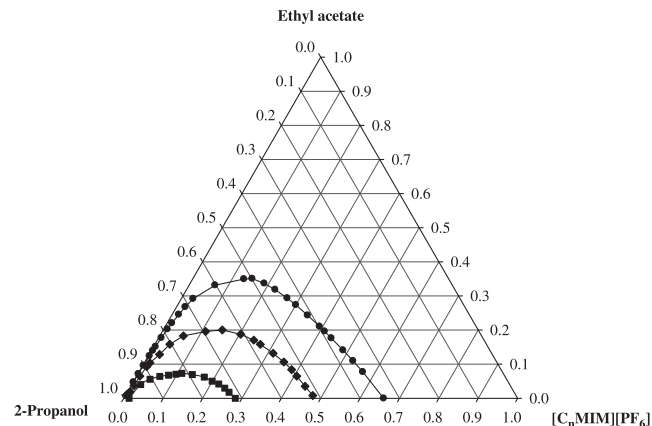


FIGURE 18. Experimental binodal curves of the ternary systems (2-propanol + ethyl acetate + [C_nMIM][PF₆]) at 298.15 K [108]: ●, [BMIM][PF₆]; ♦, [HMIM][PF₆]; and ■, [OMIM][PF₆].

2.2.2. (Alcohols + ketones) systems

Acetone and methanol are two of the most widely used solvents and their mixtures are frequent. The method for recovering volatile solvents is typically by rectification in multiplate column. Complete recovery is, however, impossible due to the formation of a minimum boiling azeotrope. For example, 2-butanone has been traditionally used as a solvent in paints and resin adhesives. Mixtures of different alcohols with this ketone forming azeotropes are very common sub-products in the coating industry.

The separation of these types of azeotropes is carried out by extractive or azeotropic distillation [109–111] using as entrainers either: 3-pentanone, acetophenone, 2,4-pentanedione, ethylacetate, or 2-butanone plus benzyl for the azeotrope methanol + acetone; amyl acetate, methyl formate, 2,2-dimethyl butane or 2,3-dimethyl butane for the azeotrope ethanol + 2-butanone; and 3-methylpentane, amyl ether or acetonitrile for the azeotrope 2-propanol + 2-butanone.

Table 8 [112–114] summarizes data using ILs for the separation of these azeotropes. Orchilles *et al.* [112] are the only group to have evaluated [EMIM][OTf] as a potential entrainer for the separation of the azeotrope methanol + acetone. They demonstrated that the addition of IL to the azeotrope produced an important salting-out effect while the azeotrope disappeared for small amounts of IL ($x_{\text{IL}} = 0.078$).

Table 9 compares [EMIM][OTf] and entrainers conventionally used to separate the methanol + acetone azeotrope. The ratio and relative volatility are not distinctly different. However, ILs are non-volatile (at the range of temperatures and pressures implemented) demonstrating significant advantages: high-boiling and thermal stable compounds, minimizing distillate contamination; and the ability to regenerate non-volatile entrainers via stripping, evaporation, drying, or crystallization.

The other alcohol + ketone azeotropes were researched by liquid–liquid extraction. A comparison between the selectivity values for the ternary systems shows that [MMIM][MeSO₄] obtains higher values than [BMIM][PF₆] for the ethanol + 2-butanone and the opposite is observed for 2-propanol + 2-butanone. All the selectivity values are higher than unity, from which it can be inferred that the extraction of the solute from the azeotropic system is indeed possible.

Figure 19 illustrates binodal curves for the systems of (ethanol + 2-butanone + [BMIM][PF₆] or [MMIM][MeSO₄]), (2-propanol + 2-butanone + [BMIM][PF₆] or [MMIM][MeSO₄]). For these two azeotropes, [BMIM][PF₆] is miscible with ketone and immiscible with alcohol. The opposite occurs for [MMIM][MeSO₄], which is miscible with alcohol and immiscible with ketone. This is a clear

TABLE 8

Azeotropes in the literature regarding the separating of alcohols + ketones. ILs used as azeotrope breakers. Solute/inert selectivity from LLE data (S^{LLE}) and acetone/methanol relative volatility from VLE data (α^{VLE}).

Azeotrope	Ionic liquids	S^{LLE} ^a	α^{VLE} ^b	Remarks	Reference
Methanol + acetone	[EMIM][OTf]		1.52	VLE data at 100 kPa	[112]
Ethanol + 2-butanone	[BMIM][PF ₆]	1.64 ^c		LLE data at 298.15 K	[113]
	[MMIM][MeSO ₄]	2.81 ^d		LLE data at 298.15 K	[114]
2-Propanol + 2-butanone	[BMIM][PF ₆]	4.14 ^c		LLE data at 298.15 K	[113]
	[MMIM][MeSO ₄]	1.53 ^d			

^a Alcohol molar fraction in organic phase $\approx$ 0.10.

^b Azeotropic composition and IL molar fraction = 0.36.

^c 2-Butanone/ethanol selectivity.

^d Ethanol/2-butanone selectivity.

TABLE 9

Ratio (volume of entrainer per volume of azeotrope) and acetone/methanol relative volatility for entrainers effective in separating of methanol + acetone azeotrope.

Entrainers	Ratio	Relative volatility
3-Pentanone ^a	3/4	1.41
Acetophenone ^a	3/5	2.27
2,4-Pentanedione ^a	3/5	1.89
Ethylacetoacetate ^a	3/5	2.17
[EMIM][OTf] ^b	3/5	1.52

^a From reference [109].

^b From reference [112].

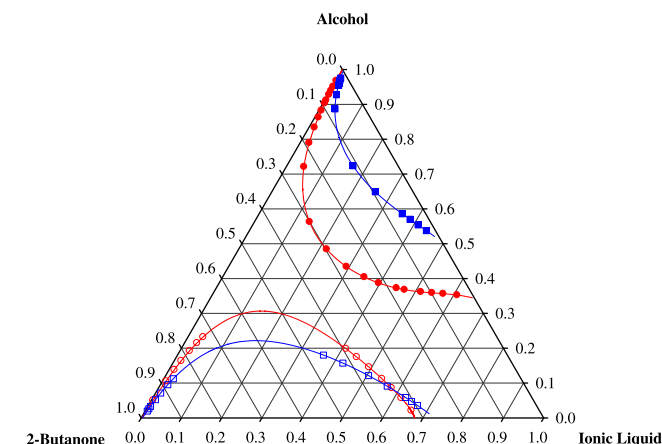


FIGURE 19. Experimental binodal curves at 298.15 K [113,114] of the ternary systems: —●— (2-propanol + 2-butanone + [BMIM][PF₆]); —■— (2-butanone + 2-propanol + [MMIM][MeSO₄]); —●— (ethanol + 2-butanone + [BMIM][PF₆]); and —■— (2-butanone + ethanol + [MMIM][MeSO₄]).

TABLE 10

Azeotropes in the literature regarding the separating of alcohols + alkanes. ILs used as azeotrope breakers. Alcohol/alkane selectivity from LLE data (S^{LLE}) at 0.01 alcohol molar fraction in organic phase.

Azeotrope	Ionic liquids	S^{LLE}	Remarks	Reference
Methanol + heptane	[OMIM][Cl]	401.0	LLE data at 298.15 K	[120]
Ethanol + hexane	[EMIM][EtSO ₄]	3631	LLE data at 298.15 K	[121]
	[HMM(EtOH)N][BF ₄]	1584	LLE data at 298.15 K	[122]
	[BMIM][MeSO ₄]	482.1	LLE data at 298.15 K	[123]
	[MMIM][MeSO ₄]	9427	LLE data at 298.15 K	[124]
	[HMIM][PF ₆]	330.7	LLE data at 298.15 K	[125]
	[OMIM][PF ₆]	115.2		
Ethanol + heptane	[EMIM][EtSO ₄]	2343	LLE data at 298.15 K	[121]
	[MMIM][MeSO ₄]	5592	LLE data at 298.15 K	[126]
	[OMIM][PF ₆]	229.8	LLE data at 298.15 K	[125]
	[BMIM][MeSO ₄]	22,326	LLE data at 298.15 K	[127]
	[HMIM][PF ₆]	2861	LLE data at 298.15 K	[128]
	[OMIM][Cl]	457.0	LLE data at 298.15 K	[120]

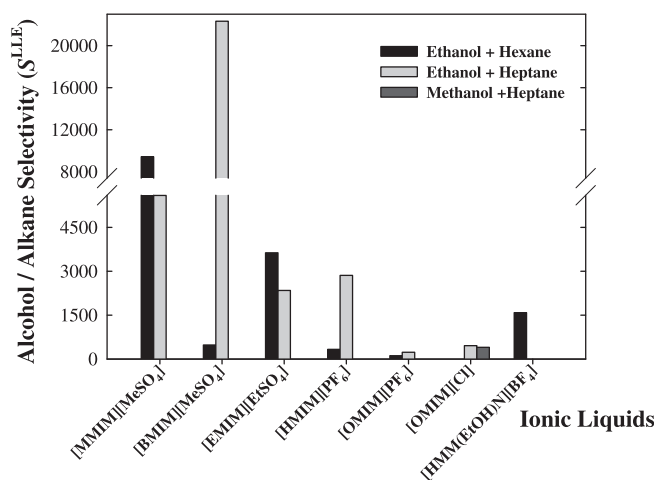


FIGURE 20. Alcohol/alkane selectivity versus ILs [120–128] used in the separation of the (alcohols + alkanes) systems at 298.15 K and alcohol molar fraction in organic phase $\approx$ 0.01.

example of the tailor-made properties of the IL: a change in the anion leads to a different extracted product. A comparison of the binodal curves shows that the immiscibility area increases when changing from ethanol to 2-propanol in the system (2-butanone + [BMIM][PF₆]), and decreases slightly when the IL is changed from [MMIM][MeSO₄] to [BMIM][PF₆] for the (ethanol + 2-butanone) system.

2.2.3. (Alcohols + aliphatic hydrocarbons) systems

The reduction of lead in gasoline has led to an increase in the number of processes under development or already in production

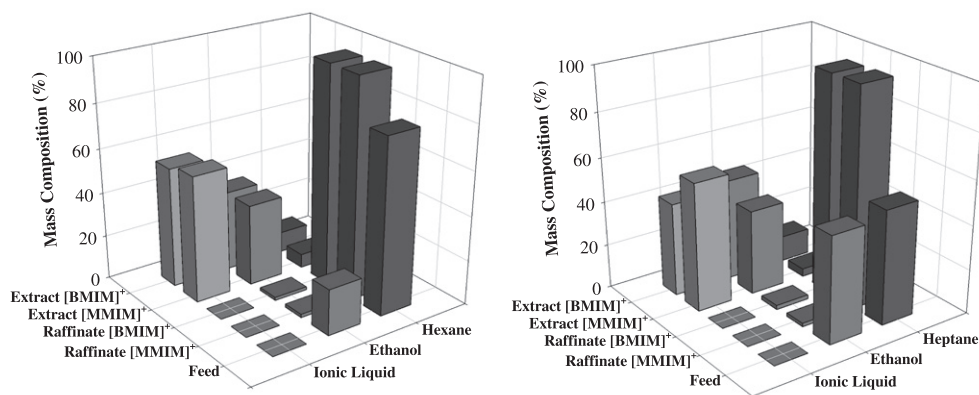


FIGURE 21. Mass composition (%) of the feed, raffinate and extract streams of the packed column under steady state conditions for ([MMIM][MeSO₄] [124,126] and [BMIM][MeSO₄] [123,127].

TABLE 11

Extraction efficiencies (from equation (6)) for the separation of the azeotrope ethanol + alkanes in the packed column [123,124,126,127,129].

Azeotrope	[MMIM][MeSO ₄]	[BMIM][MeSO ₄]
Ethanol + hexane	1.81	1.14
Ethanol + heptane	1.26	1.13

[115] in which alkanols and alkanes are brought together to produce oxygenated additives for gasolines. The azeotropes of either hexane or heptane with methanol or ethanol are published due to difficulties in separating these compounds.

The liquid–liquid separation leads to an environmentally friendly extraction process of these azeotropes as an alternative to the more common processes such as azeotropic distillation [116,117], pervaporation [118], and reverse osmosis [119]. Table 10 [120–128] evaluates IL azeotrope breakers for alkanols + alkanes. Figure 20 compares alcohol/alkane selectivity values for the studied ternary systems for the ILs listed in table 10. It can be concluded that a short alkyl side chain on the imidazolium cation increases selectivity, favoring ethanol/alkane separation.

Based on the analysis of the ILs extraction capacities for ethanol + alkanes (hexane or heptane) systems, ILs with the methyl sulfate anion ([MMIM][MeSO₄] and [BMIM][MeSO₄]) were included in a lab-scale extraction process incorporating a solvent recycling stage [123,124,126,127,129]. These selectivities are the highest reported in the literature for azeotropic separation in general using liquid–liquid extraction. The (liquid + liquid) equilibria data allowed researchers to identify theoretically appropriate operating conditions for countercurrent continuous extraction process at room temperature including a solvent recycling stage. Simulation techniques then optimized operational conditions. Experiments with a laboratory-scale packed column under steady-state conditions achieved a raffinate purity of over 98 wt%.

The lab-scale experiments also confirmed the possibility of ready online recovery of selected ILs, by evaporating the organic

compounds from the extract to regenerate the ILs used in this extraction process. Visible inspection of the NMR spectra and physical properties of both the original IL and the regenerated product showed no differences between the two samples. This means that no degradation of the IL after repeated recycling was confirmed.

The performance of the experimental column exceeds the theoretical (from (liquid + liquid) equilibria data) and simulation-based expectations, certainly due to the superior mixing of the feed and the solvent. Figure 21 compares the compositions of the initial feed and the two outlet streams of the countercurrent packed column for the two ILs ([MMIM][MeSO₄] and [BMIM][MeSO₄]). The results clearly indicate that [MMIM][MeSO₄] has less alkane in the extract stream. In order to emphasize the two ILs for extraction processes in the packed column, the extraction efficiency, E^{EP} , was calculated (from equation (6)). This parameter indicates the ability of the ILs to remove solute (ethanol) from the azeotropes (ethanol + alkane) in the extraction column. Table 11 lists the extraction efficiencies of the ILs in the separation of the ethanol + alkanes (hexane and heptane) azeotropes. The values obtained for [MMIM][MeSO₄] are higher than those for [BMIM][MeSO₄]. Moreover, [MMIM][MeSO₄] has both lower viscosity and synthesized cost.

2.2.4. (Alcohols + halogenated) systems

Nowadays, great importance is given to the separation of organic waste composed mainly of chloroform and methanol, a widely used effective solvent mixture for the extraction of bioactive substances from biological sources in the pharmaceutical and biotechnological industry [130–132]. This binary system exhibits a minimum boiling temperature azeotrope $x_{\text{Chloroform}} = 0.648$ at 328.15 K and atmospheric pressure (table 1). Then, non-conventional distillation processes as extractive or azeotropic distillation are required to separate both components. In addition to its industrial importance, theoretical issues regarding the interesting behavior verified in systems containing chloroform and lower aliphatic alcohols were first addressed by Scatchard *et al.* [133] through VLE measurements.

TABLE 12

Azeotropes in the literature regarding the separating of alcohols + halogenated. ILs used as azeotrope breakers. Alcohol/chloroform selectivity from LLE data (S^{LLE}) and chloroform/alcohol relative volatility from VLE data (α^{VLE}) at azeotropic composition.

Azeotrope	Ionic liquids	S^{LLE} ^a	α^{VLE}	Remarks	Reference
Methanol + chloroform	[EMIM][OTf]	5.42	1.62 ^b	VLE data at 100 kPa + LLE data at 293.15 K	[134]
Ethanol + chloroform	[EMIM][OTf]	3.62	1.82 ^c	VLE data at 100 kPa + LLE data at 293.15 K	[135]

^a Alcohol molar fraction in organic phase ≈ 0.01 .

^b Azeotropic composition and IL molar fraction = 0.35.

^c Azeotropic composition and IL molar fraction = 0.32.

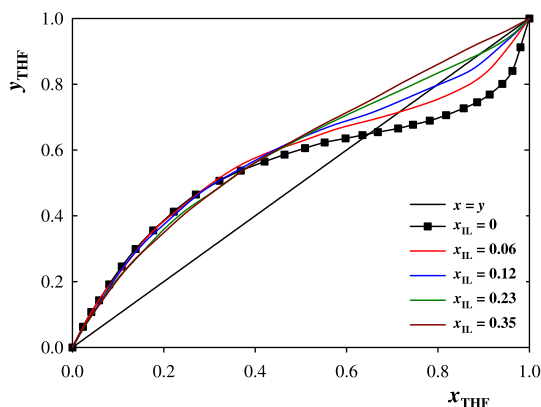


FIGURE 22. Influence of [EMIM][OTf] on the VLE of (chloroform + methanol) system at 100 kPa for different IL concentrations [134].

The ILs reported in table 12 [134,135] were used as azeotrope breakers for the (alcohols + chloroform) systems using extractive distillation and liquid–liquid extraction.

Orchilles and co-workers [134,135] presented the effect of [EMIM][OTf] on the VLE for the systems (chloroform + methanol) and (chloroform + ethanol). Both systems exhibit, at atmospheric pressure, a minimum boiling temperature azeotrope at $x_{\text{Chloroform}} = 0.648$ for the methanol system and $x_{\text{Chloroform}} = 0.841$ for the ethanol system (table 1). For these systems, inorganic salts such as lithium chloride [136,137], calcium chloride [137,138], calcium nitrate [139], potassium acetate [137], and sodium iodide [139,140] have been used, rather unsuccessfully, as entrainers to break the azeotrope mostly due to the low solubility of these salts at large chloroform concentrations. These authors demonstrated the capacity of [EMIM][OTf] to break both azeotropes and its aptness as entrainer in extractive distillation is clearly elucidated in figure 22.

Analysing the displacement of azeotropic point depicted in figure 22 as a function of [EMIM][OTf] concentration, it is readily observed that, for chloroform + methanol at a IL composition of 0.23, the azeotrope has not yet disappeared, but for the chloroform + ethanol system (supporting information figure S9) at a IL composition of 0.21 the azeotrope has done so. This is explained by the fact that the chloroform relative volatility is higher in the (chloroform + ethanol) system than in the (chloroform + methanol) system. Furthermore, since methanol and ethanol possess similar dipole moments, it is obvious that the (chloroform + methanol) system will need a greater electrolyte concentration to the azeotrope.

Both methanol and ethanol are more polar than chloroform. Hence, the latter is supposed to be salted-out from the mixture over the whole range of liquid concentration. However, figure 22 demonstrates that the salting-out effect is only verified for chloroform compositions higher than 0.4 for the system with methanol, and higher than 0.5 for the system with ethanol (supporting information figure S9). On the other hand, a salting-in effect occurs at lower compositions. [EMIM][OTf] produces a cross-over effect [141], a fact clearly demonstrated in figure 23 by the selectivity values for both salting-in and salting-out regions in (chloroform + methanol + [EMIM][OTf]) and (chloroform + ethanol + [EMIM][OTf]) systems.

Orchilles and co-workers [134,135] also researched the separation of the alcohol + chloroform azeotropes by liquid–liquid extraction using [EMIM][OTf] as solvent. These systems present small immiscibility regions, increasing with temperature at the highest chloroform compositions. The highest alkanol/chloroform selectivities for [EMIM][OTf] were obtained at low concentration of alcohol in organic phase (supporting information figure S10).

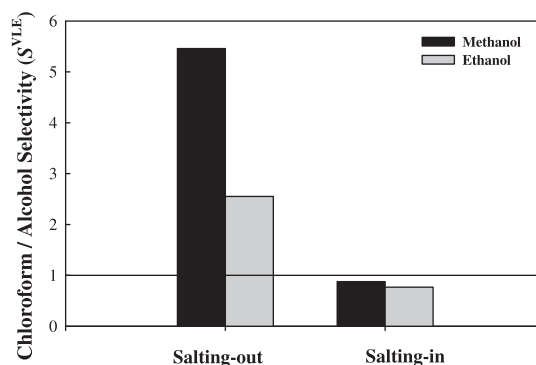


FIGURE 23. Influence of the [EMIM][OTf] crossover effect on the selectivity for both salting-out ($x_{\text{Chloroform}} = 0.9407$) and salting-in ($x_{\text{Chloroform}} = 0.1081$) regions [134,135]. The straight horizontal line represents the selectivity value, equal to unity, when the relative volatility is equal with and without the entrainer.

2.3. (Aromatic + aliphatic hydrocarbons) systems

Several processes [142] for the separation of aromatic and aliphatic hydrocarbon mixtures in reformat gasoline distilled from crude have been developed: liquid extraction for aromatic contents from 20 to 65 wt%, extractive distillation for 65 to 90 wt%, and azeotropic distillation for aromatic content higher than 90 wt%. These products are the most widespread and important raw materials in plastic and synthetic fiber manufacture [143]. Aromatics are obtained almost exclusively from fuels, coal, and oil; however, the most economically feasible process is to remove them from fuels.

Typical solvents used in the separation of (aromatic + aliphatic) mixtures, through liquid–liquid extraction, are sulfolane [144–146], N-methylpyrrolidone [144,147], N-formylmorpholine [146,148], ethylene glycols [149,150], and propylene carbonate [151]. When these conventional solvents are used, additional distillation steps are required to separate the solvent from both the extract and raffinate phases and to purify the solvent, increasing separation costs. The costs of regeneration of sulfolane are high, because, in the current process, sulfolane, which has a boiling temperature of 287.3 °C, is taken as overhead from the regenerator and returned to the bottom of the aromatics stripper as a vapor [152].

Most ethylene cracker feeds contain 10 to 25 wt% aromatic components, depending on the source of the feed (naphtha or gas condensate). The aromatic compounds are not converted into olefins, and even small amounts are formed during the process in the cracker furnaces. Therefore, these compounds partly occupy the furnace capacity, putting an extra load on the separation section of the stream containing C_5 – C_{10} aliphatic compounds. If a significant part of the aromatic compounds present in the feed to the crackers could be separated upstream, it would offer several advantages: higher capacity, higher thermal efficiency, and less fouling [153]. The improved margin for the removal of 10 wt% aromatic hydrocarbons from the feed to the naphtha cracker would represent a savings of around € 20/t of feed or € 48 million per year for a cracker with a feed capacity of 300 t/h [153]. Although the separation of aromatic and aliphatic hydrocarbons after the furnace section is current practice, no suitable technology yet exists for separating aromatic compounds from the feed to cracker plants [153]. Different ILs for the separation of (aromatic + aliphatic hydrocarbons) azeotropic systems were therefore tested to $w_{\text{Aromatic}} = 0.10$ in organic phase.

Table 13 [30,65,153–179] lists the aromatic/aliphatic hydrocarbons azeotropes using ILs as azeotrope breakers. These research data study the azeotropic separation by extractive distillation (3 papers), liquid–liquid extraction (26 papers) and supported liquid membrane (4 papers).

TABLE 13

Azeotropes in the literature the separating of aromatic + aliphatics hydrocarbons. ILs used as azeotrope breakers. Aromatic/aliphatic selectivity from LLE data (S^{LLE}). Aliphatic/aromatic relative volatility from VLE data (α^{VLE}). Aromatic/aliphatic selectivity from membranes data (S^{MEM}) and aromatic/aliphatic separation factor from membranes data (α^{MEM}).

Azeotrope	Ionic liquids	S^{LLE} ^a	α^{VLE} ^b	S^{MEM}	α^{MEM}	Remarks	Reference
Benzene + hexane	[HMIM][NTf ₂]		9.49			VLE data at 322.8 K: headspace gas chromatography	[30]
	[OMIM][NTf ₂]		7.53				
	[EMIM][N(CN) ₂]	30.10				LLE data at 303.15 K	[153]
	[EMIM][SCN]	114.3					
	[BMIM][N(CN) ₂]	30.50					
	[BMIM][C(CN) ₃]	32.30					
	[BMIM][SCN]	55.40					
	[HMIM][SCN]	20.90					
	[3MBpy][BF ₄]	36.20					
	[3MBpy][N(CN) ₂]	35.30					
	[3MBpy][C(CN) ₃]	34.80					
	[3MBpy][B(CN) ₄]	27.00					
	[4MBpy][N(CN) ₂]	34.30					
	[4MBpy][SCN]	45.20					
	[BMpyr][NTf ₂]	20.70					
	[BMpyr][N(CN) ₂]	15.40					
	[BMpyr][SCN]	17.30					
	[MOON][NTf ₂]	1.80					
	[EEES][NTf ₂]	32.20					
	[3MBpy][N(CN) ₂]	48.88				LLE data at 303.15 K	[154]
	[BMIM][MeSO ₄]	57.99				LLE data at 298.15 K	[155]
	[EPy][EtSO ₄]	62.30				LLE data at 298.15 K	[156]
	[EMIM][EtSO ₄]	48.55				LLE data at 298.15 K	[157]
	[EMPy][EtSO ₄]	26.12				LLE data at 298.15 K	[158]
	[EMIM][EtSO ₄]	87.68				LLE data at 298.15 K	[159]
	[EPy][NTf ₂]	27.70				LLE data at 313.15 K	[160]
	[MMM(EtOH)N][NTf ₂]	41.90					
	[P _{6 6 14}][NTf ₂]	1.07				LLE data at 298.15 K	[161]
	[EMIM][NTf ₂]	27.90				LLE data at 298.15 K	[162]
	[BMIM][NTf ₂]	15.89				LLE data at 298.15 K	[163]
	[OMIM][NTf ₂]	7.35					
	[DMIM][NTf ₂]	4.90					
	[DoMIM][NTf ₂]	3.54					
	[EMM(EtOH)N][NTf ₂]	20.51				LLE data at 298.15 K	[164]
	[4MBpy][BF ₄]	55.01				LLE data at 313.15 K	[165]
Benzene + heptane	[EMIM][EtSO ₄]	71.66				LLE data at 298.15 K	[157]
	[BMIM][BF ₄]	110.0				LLE data at 298.15 K	[166]
	[BMIM][SCN]	95.63					
	[MMIM][MePO ₃]	843.2					
	[BMpyr][NTf ₂]	17.91				LLE data at 298.15 K	[167]
	[EPy][EtSO ₄]	67.27				LLE data at 298.15 K	[156]
	[EMPy][EtSO ₄]	71.60				LLE data at 298.15 K	[158]
	[BMIM][Cl] – 2.0 AlCl ₃	80.00				LLE data at 293.15 K	[168]
	[MMMNH][Cl] – 2.0 AlCl ₃	38.00					
	[EEENH][Cl] – 2.0 AlCl ₃	40.20					
	[EEENH][Cl] – 1.5 AlCl ₃	32.00					
	[EEENH][Cl] – 1.2 AlCl ₃	28.00					
	[EMIM][OCSO ₄]	2.71				LLE data at 298.15 K	[169]
	[OMIM][Me(EtO) ₂ SO ₄]	0.63					
	[HMIM][BF ₄]	8.35				LLE data at 298.15 K	[170]
	[HMIM][PF ₆]	11.20					
	[OMIM][Cl]	6.06				LLE data at 298.15 K	[171]
	[BMIM][PF ₆]			67.00		Supported liquid membranes: permeation	[172]
	[HMIM][PF ₆]			16.00			
	[OMIM][PF ₆]			7.50			
	[EEM(MeOEt)N][NTf ₂]			7.20			
Benzene + cyclohexane	[OMIM][NTf ₂]		3.93			VLE data at 322.8 K: headspace gas chromatography	[30]
	[HMIM][NTf ₂]		5.04			VLE data at 332 K	[65]
	[BMpyr][NTf ₂]		7.75				
	[EMPy][EtSO ₄]	26.97				LLE data at 298.15 K	[173]
	[4MBpy][BF ₄]	28.25				LLE data at 303.15 K	[174]
	[MIM][(Bu) ₂ PO ₄]	1.90				LLE data at 298.15 K	[175]
	[MMIM][(Me) ₂ PO ₄]	3.60				LLE data at 298.15 K	[176]
	[EMIM][(Et) ₂ PO ₄]	4.30					
	[BMIM][PF ₆]			11.60		Supported Liquid Membranes: Pervaporation	[177]
	[HMIM][PF ₆]			10.30			
	[OMIM][PF ₆]			2.50			
	[EEM(MeOEt)N][NTf ₂]			10.90			
	[EEM(MeOEt)N][BF ₄]			47.10			
	[BMIM][PF ₆]				19.70	Supported liquid membranes: vapor permeation	[178]
	[HMIM][PF ₆]				10.80		

(continued on next page)

TABLE 13 (continued)

Azeotrope	Ionic liquids	S^{LLE} ^a	α^{VLE} ^b	S^{MEM}	α^{MEM}	Remarks	Reference
Pyridine + heptane	[OMIM][PF ₆]				5.90	Supported liquid membranes: permeation	[179]
	[EEM(MeOEt)N][NTf ₂]				6.92		
	[EEM(MeOEt)N][BF ₄]				185.0		
	[BMIM][PF ₆]			9.00			
	[HMIM][PF ₆]			7.60			
	[OMIM][PF ₆]			4.90			
	[EEM(MeOEt)N][NTf ₂]			3.80			
	[EEM(MeOEt)N][BF ₄]			8.60			

^a Aromatic mass fraction in organic phase ≈ 0.10 .

^b Azeotropic composition and IL molar fraction ≈ 0.81 .

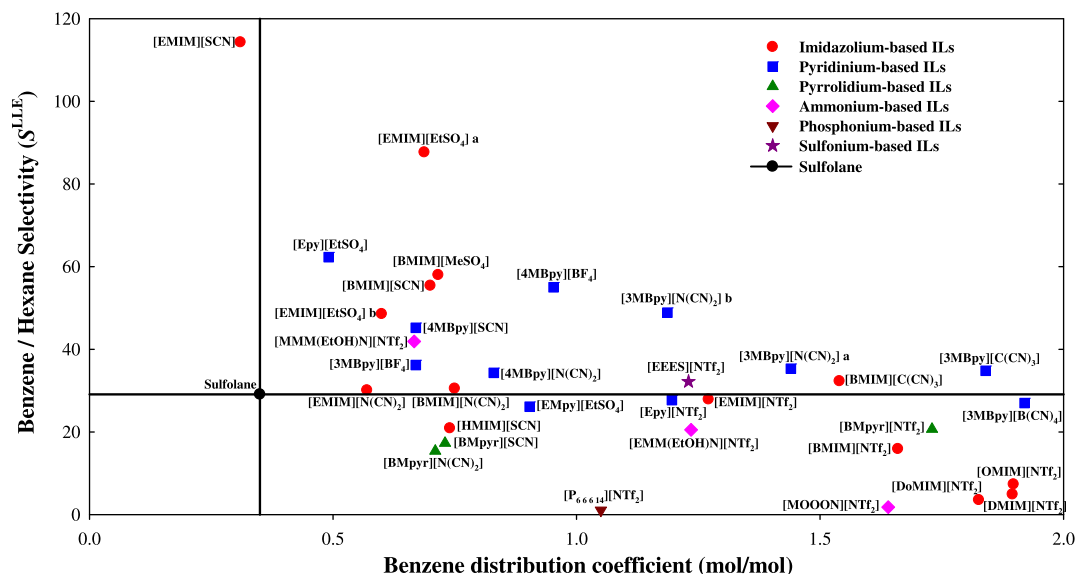


FIGURE 24. Benzene/hexane selectivity versus benzene distribution coefficient for the ILs used in the separation of (benzene + hexane) [153–155] at ≈ 298.15 K and benzene mass fraction in organic phase ≈ 0.10 . [EMIM][EtSO₄] (a) from reference [159] and [EMIM][EtSO₄] (b) from reference [157]; [3MBpy][EtSO₄] (a) from reference [153] and [3MBpy][EtSO₄] (b) from reference [154].

2.3.1. (Benzene + hexane) system

While most research has focussed on liquid–liquid extraction, Mokhtarani *et al.* [30] studied the separation the azeotrope (benzene + hexane) by extractive distillation. These results showed that the addition of [HMIM][NTf₂] and [OMIM][NTf₂] as entrainers improves separation.

The system (benzene + hexane) is representative of aromatic/aliphatic hydrocarbon separation by liquid–liquid extraction. IL criteria were: a comparable or higher benzene distribution coefficient and benzene/hexane selectivity than sulfolane (0.35 mol/mol and 29.1 at 303.15 K and $w_{\text{Aromatic}} = 0.10$ in organic phase, respectively) [153]. Temperature has been shown to have little influence on the phase equilibria of this kind of system [160,162]. Figure 24 plots benzene/hexane selectivity as a function of the distribution coefficient (mol/mol) of benzene compared with the sulfolane values. The most suitable ILs for the separation of benzene and hexane are: [3MBpy][C(CN)₃] with a distribution coefficient of 1.84 (mol/mol) and a selectivity of 34.8; [BMIM][C(CN)₃] with a distribution coefficient of 1.54 (mol/mol) and a selectivity of 32.3; [3MBpy][N(CN)₂] with a distribution coefficient of 1.44 (mol/mol) and a selectivity of 35.3 [153] or a distribution coefficient of 1.19 (mol/mol) and a selectivity of 48.9 [154]; and [3MBpy][B(CN)₄] with a distribution coefficient of 1.92 (mol/mol) and a selectivity of 27.0. All values are higher than those obtained with sulfolane.

2.3.1.1. Effect of the alkyl chain length. Figure 25 illustrates the evolution of the benzene/hexane selectivity and benzene distribution coefficient as a function of the cation alkyl chain length in the IL [C_nMIM][NTf₂]. The use of the IL with the shortest alkyl chain, $n = 2$ ([EMIM][NTf₂]), leads to the highest selectivity values. For the distribution coefficient, however, [EMIM][NTf₂] has the lowest values, which rise with moderate increases in the length of the alkyl chain of the IL. The increment in the distribution coefficient is smaller than the decline in selectivity values when the IL chain of the IL lengthens.

Benzene is known to interact with ILs such as [EMIM][NTf₂] to form an inclusion compound, [EMIM][NTf₂]-C₆H₆ [180]. The resultant stability is derived from the anions clustering around the benzene and derives from interactions between the benzene hydrogen atoms and the oxygen atoms of the sulfonyl groups in the anions [180]. The compound decreases benzene–hexane (azeotrope) interactions, facilitating separation. Cation benzene interactions are mainly due to π - π stacking which promotes quadrupolar benzene molecule associates with the anion and the imidazolium part of the cation.

The distribution coefficients (figure 25) at low benzene concentrations in organic phase ($w_{\text{Aromatic}} = 0.10$) presented values in the following order: C₁₂ < C₁₀ $\approx$ C₈ > C₄ > C₂. A possible explanation is that hexane is moderately soluble in [DoMIM][NTf₂] ($n = 12$) [163]. The increasing hexane concentration in the IL as the chain

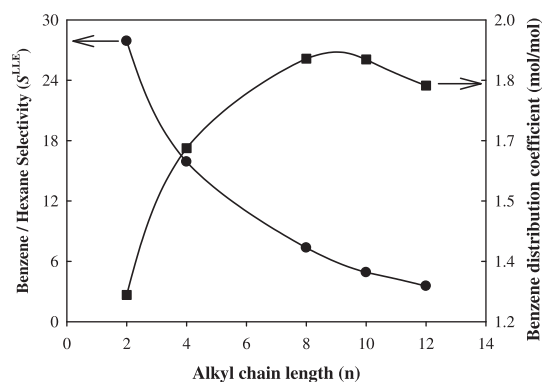


FIGURE 25. Effect of cation alkyl chain length on benzene/hexane selectivity and benzene distribution coefficient in ILs $[C_n\text{MIM}][\text{NTf}_2]$ [162,163] at 298.15 K and benzene mass fraction in organic phase ≈ 0.10 .

length increases lowers the distribution ratio [163]. The best distribution ratios occur for ILs that are immiscible with benzene. Short chain ILs, such as $[\text{EMIM}][\text{NTf}_2]$, can dissolve benzene quite well, but, for hexane the anion–cation interactions would be disrupted if they were to mix. For long chain $[C_n\text{MIM}][\text{NTf}_2]$ ILs, hexane can be accommodated in the hydrophobic region of the alkyl side chains and hence its solubility in the IL phase improves [163]. Benzene can also be accommodated in these nonpolar regions, and consequently its solubility increases to such an extent that it is completely miscible with $[\text{DoMIM}][\text{NTf}_2]$ [163].

2.3.1.2. Effect of the nature of the anion. Figure 26 depicts the benzene/hexane selectivity regarding the separation of the azeotrope benzene + hexane using various ILs with differing anions as solvents. These ILs are based on the $[\text{EMIM}]^+$ and $[\text{BMIM}]^+$ cations which were combined with six different anions.

The effect of the anion on selectivity follows the trend for the $[\text{EMIM}]^+$ cation: $[\text{SCN}]^- > [\text{EtSO}_4]^- > [\text{N}(\text{CN})_2]^- \approx [\text{NTf}_2]^-$; and for the $[\text{BMIM}]^+$ cation: $[\text{MeSO}_4]^- \approx [\text{SCN}]^- > [\text{C}(\text{CN})_3]^- \approx [\text{N}(\text{CN})_2]^- > [\text{NTf}_2]^-$. Generally, anions with delocalized charge and with small molar volume seem to be efficient IL solvents. Especially $[\text{MeSO}_4]^-$, $[\text{SCN}]^-$, and $[\text{EtSO}_4]^-$ are the best candidates for forming an effective solvent for this liquid–liquid extraction. Anions with delocalized charges interact less with cations, so benzene interactions with cations are favored. Furthermore, with low molecular weights, the number of IL molecules in the vicinity of the benzene molecule is high because of its low molar volume. The result is increased benzene selectivity with these ILs.

2.3.1.3. Effect of the nature of the cation. Figure 27 depicts benzene/hexane selectivity as a function of the nature of the cation on the separation of the azeotrope benzene + hexane, using as solvent various ILs with the dicyanamide or bis(trifluoromethylsulfonyl)imide anions. These selectivities at low benzene concentrations in organic phase ($w_{\text{Aromatic}} = 0.10$) present values in the following order: $[\text{3MBpy}]^+ > [\text{4MBpy}]^+ > [\text{EMIM}]^+ \approx [\text{BMIM}]^+ > [\text{MBpyr}]^+$ for the $[\text{N}(\text{CN})_2]^-$ anion; and $[\text{MMM}(\text{EtOH})\text{N}]^+ > [\text{EEES}]^+ > [\text{Epy}]^+ \approx [\text{EMIM}]^+ > [\text{MBpyr}]^+ \approx [\text{EMM}(\text{EtOH})\text{N}]^+ > [\text{BMI-M}]^+ > [\text{MOOON}]^+ \approx [\text{P}_{6,6,6,14}]^+$ for the $[\text{NTf}_2]^-$ anion. Again, the reported benzene/hexane selectivities in the abovementioned ILs decrease with increasing alkyl chain length of the $[C_n\text{MIM}]^+$ cation. This behavior is observed in ammonium-based, imidazolium-based and pyridinium-based ILs.

For the same $[\text{N}(\text{CN})_2]^-$ anion and similar alkyl chain length, pyridinium ILs have higher selectivities than imidazolium ILs. This can be probably attributed to the higher aromatic character of pyridinium. This behavior is also observed for the $[\text{NTf}_2]^-$ anion.

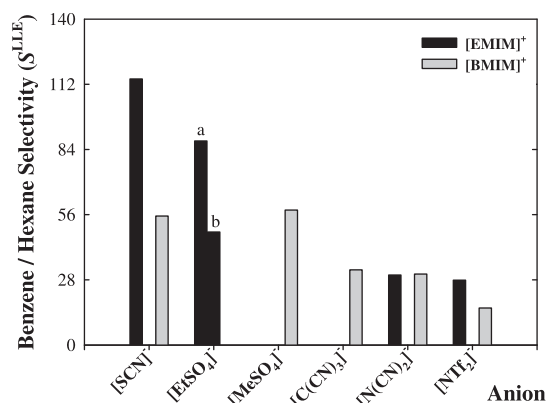


FIGURE 26. Effect of the anion nature of ILs with the $[\text{EMIM}]^+$ or $[\text{BMIM}]^+$ cation on the benzene/hexane selectivity for the benzene + hexane azeotrope [153,155,157, 159,162,163] at benzene mass fraction in organic phase ≈ 0.10 (for $[\text{EtSO}_4]^-$ anion: (a) from reference [159] and (b) from reference [157]).

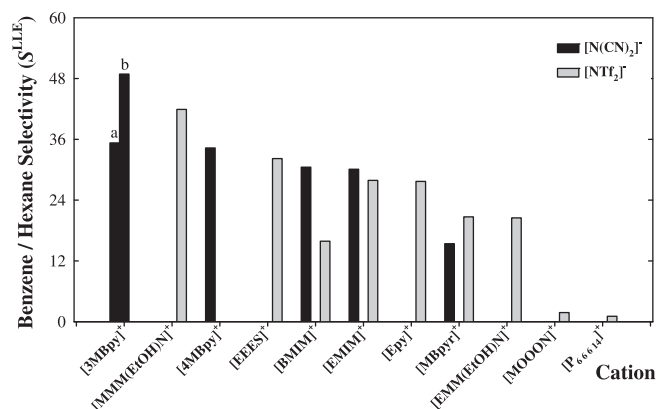


FIGURE 27. Effect of the cation nature of ILs with the $[\text{N}(\text{CN})_2]^-$ or $[\text{NTf}_2]^-$ anion on the benzene/hexane selectivity for the (benzene + hexane) azeotrope [153,154,160–164] at benzene mass fraction in organic phase ≈ 0.10 (for $[\text{3MBpy}]^+$ cation: (a) from reference [153] and (b) from reference [154]).

Moreover, in the case of the $[\text{MMM}(\text{EtOH})\text{N}]^+$ cation system, its hydroxyl group is likely to favorably interact with the aromatic ring of benzene, while a cation– π interaction may also exist [160].

2.3.2. (Benzene + heptane) system

Subsequent to the (benzene + hexane) system, benzene + heptane can be used to model aromatic/aliphatic hydrocarbons separations. The benzene distribution coefficient for sulfolane is 0.46 mol/mol and the benzene/heptane selectivity is 22.2 at 303.15 K and $w_{\text{Aromatic}} = 0.10$ in organic phase [181]. The IL selection criteria were those for where the benzene distribution coefficient is greater than 0.46 mol/mol and benzene/hexane selectivity greater than 22.2, at $T \approx 298.15$ K and at $w_{\text{Benzene}} = 0.10$ in organic phase (comparison between ILs and sulfolane in supporting information figure S11). Room temperature was considered as the reference due to the research tendency to determine the (liquid + liquid) equilibria of ternary systems at this temperature, given that all compounds and their mixtures are liquid at such temperature.

Table 13 lists the ILs suitable for benzene/heptane separation. The chloroaluminate IL $[\text{BMIM}][\text{Cl}]-2.0 \text{ AlCl}_3$ shows the highest benzene distribution coefficient and the highest benzene/heptane selectivity (supporting information figure S11), but this IL is not suitable for industrial processes due to its reaction with water

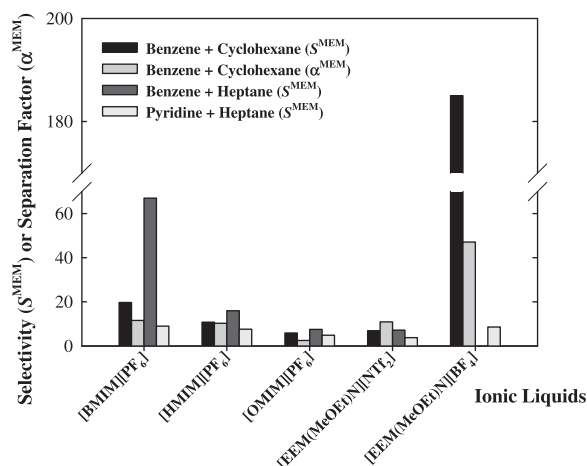


FIGURE 28. Selectivity or separation factor versus ILs used in the separation of the (aromatics + aliphatic) systems by supported liquid membranes [172,177–179].

[153]. The ILs [MMNH][Cl]–2.0 AlCl₃ and [EEENH][Cl]–2.0 AlCl₃ (or –1.5 AlCl₃ or –1.2 AlCl₃) are not suitable for the same reason (protic ILs). In the case of [BMIM][BF₄] IL, with its high selectivity value, the anion [BF₄][–] has been commonly investigated and considered historically to be one of the most important anion families. However, this anion can undergo hydrolysis producing hydrofluoric acid when in contact with water and at high temperatures [182,183]. Consequently, its industrial application has been restricted to those processes under water-free conditions and moderate temperatures.

The most suitable ILs for the separation of benzene and heptane are: [BMIM][SCN] with a distribution coefficient of 1.03 (mol/mol) and a selectivity of 95.6; [EMpy][EtSO₄] with a distribution coefficient of 0.80 (mol/mol) and a selectivity of 71.6; [EMIM][EtSO₄] with a distribution coefficient of 0.56 (mol/mol) and a selectivity of 71.7; and [BMpyr][NTf₂] with a distribution coefficient of 1.25 (mol/mol) and a selectivity of 17.9. All these values are higher than those obtained with sulfolane for the separation of the azeotrope benzene + heptane.

2.3.3. Other (aromatic + aliphatic hydrocarbons) systems

The azeotrope (benzene + cyclohexane) system was studied using three separation processes: liquid–liquid extraction, extractive distillation and supported liquid membranes. Table 13

collects the ionic liquids used as azeotrope breakers for this system together with the values of the parameters for each process with the aim of evaluating separation efficiency. The best extractive distillation entrainer IL is [BMpyr][NTf₂]. In the case of liquid–liquid extraction, [EMpy][EtSO₄] and [4MBpy][BF₄] present the highest values of benzene/cyclohexane selectivity.

Figure 28 demonstrates the results for the separation of the benzene + cyclohexane or heptane and pyridine + heptane using supported liquid membranes. These data show that the use of ILs as a membrane solvent enables this technique to be used for the separation of aromatic and aliphatic hydrocarbons. Aromatic hydrocarbons were successfully transported through the membrane. The hydrophilic liquid membrane that used [EEM(MeO–Et)N][BF₄] led to the highest separation factor for the azeotrope benzene + cyclohexane, 185 for 53 wt% benzene obtained by vapor permeation, which is superior to the reported values for pervaporation (selectivity of 47.1 for the mixed solution). Other good results were obtained for the azeotrope benzene + heptane. The maximum selectivity to heptane resulted from the use of benzene in the aromatic permeation and [BMIM][PF₆] in the liquid membrane phase (selectivity of 67.0 for the mixed solution).

2.4. Other azeotropes

2.4.1. (Methanol + dimethyl carbonate) system

Dimethyl carbonate has been used as an effective carbonylation agent to substitute phosgene in manufacturing polycarbonate plastics and as a useful methylation agent to replace the dimethyl sulfate [184]. Moreover, dimethyl carbonate has been successfully manufactured using carbon dioxide, the greenhouse gas, and methanol as raw materials according to a zero-discharge process co-producing propylene glycol [185]. As a result, with respect to the carbonylation usage of dimethyl carbonate, phosgene is simply replaced with carbon dioxide, which may be one of the most significant improvements in the chemical industry. Dimethyl carbonate is prepared from the catalytic oxidative carbonylation of methanol and forms an azeotrope with methanol. The separation of this azeotrope has been investigated by extractive distillation [186] and/or by the use of membranes [187,188].

Kim *et al.* [64] reports the influence of [EMIM][EtSO₄] and [BMIM][BF₄] on the separation of the azeotrope dimethyl carbonate + methanol (table 14, [64,189–193]). Both ILs were effective in breaking the azeotropic point, by increasing the mole fraction of methanol in the vapor phase. The azeotrope disappeared at

TABLE 14

Azeotropes in the literature regarding the separating of other azeotropes. ILs used as azeotrope breakers. Selectivity from LLE data (S^{LLE}) and relative volatility from VLE data (α^{VLE}).

Azeotrope	Ionic liquids	S^{LLE} ^a	α^{VLE}	Remarks	Reference
Methanol + dimethyl carbonate	[EMIM][EtSO ₄]		2.05 ^b	VLE data at 333.15 K: headspace gas chromatography	[64]
	[BMIM][BF ₄]		1.71 ^b		
Ethanol + ethyl tert-butyl ether	[EMIM][EtSO ₄]	251.7 ^c	3.42 ^d	VLE data at 101.32 kPa + LLE data at 298.15 K	[189]
	[EMIM][OTf]	47.30 ^c		LLE data at 298.15 K	
	[EMIM][MeSO ₃]	1095 ^c		LLE data at 298.15 K	[191]
	[BMIM][OTf]	20.28 ^c			
Hexane + ethyl acetate	[HMIM][PF ₆]	186.2 ^e		LLE data at 298.15 K	[192]
	[OMIM][PF ₆]	83.29 ^e			
Cyclohexane + 2-butanone	[HMIM][PF ₆]	114.9 ^f		LLE data at 298.15 K	[193]
	[OMIM][PF ₆]	55.30 ^f			

^a Solute molar fraction in organic phase $\approx$ 0.01.

^b Methanol/dimethyl carbonate relative volatility at azeotropic composition and IL molar fraction = 0.20.

^c Solute = ethanol.

^d Ethyl tert-butyl ether/ethanol relative volatility at azeotropic composition and IL molar fraction = 0.31.

^e Solute = ethyl acetate.

^f Solute = 2-butanone.

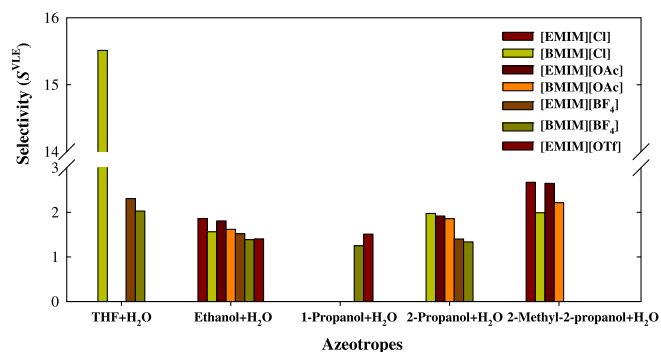


FIGURE 29. Selectivity (S^{VLE}) for the ILs [27,34,37,62,63,69,71] used as entrainers in the aqueous azeotropes separated by extractive distillation at THF or alcohol molar fraction ≈ 0.95 and $x_{\text{IL}} \approx 0.10$.

$x_{\text{IL}} = 0.1$. There is no significant difference between the two ILs, but [EMIM][EtSO₄] showed a slightly better salting-out effect.

2.4.2. (Ethanol + ethyl tert-butyl ether) system

The production of tertiary ethers at an industrial scale is mainly carried out by the reaction of an olefine and an alcohol. Specifically, ethyl tert-butyl ether (ETBE) is obtained by the reaction of isobutene and ethanol. Due to the required excess of ethanol in the reactor, the output stream of the unit is a mixture of ETBE and unreacted ethanol. Although both compounds have an anti-knock effect, blending properties of ETBE are largely better than those of ethanol [190]. So far, water has been the solvent most commonly used in purification stages of ETBE by liquid extraction [190], due to its low cost and physical properties.

Arce *et al.* [189–191] demonstrate the ability of the ILs based on imidazolium cations to act as extraction solvents for liquid–liquid extraction and as azeotrope breakers for extractive distillation (table 14) in order to separate azeotrope ETBE + ethanol, thus purifying the tertiary ether, which is the most common additive for improving gasoline octane indexes. A comparison between ethanol/ETBE selectivity values for all the ILs used as solvents in liquid–liquid extraction (supporting information figure S12) shows that the use of [EMIM][MeSO₃] as the leads to the highest selectivity values. Moreover, the data depicted in figure S12 indicates that the fluorination of the anion plays a negative role in the capability of the IL to purify the ether by liquid extraction.

2.4.3. (Hexane + ethyl acetate) and (cyclohexane + 2-butanone) systems

Pereiro *et al.* [192,193] chose the ILs [HMIM][PF₆] and [OMIM][PF₆] for the separation of the azeotropic systems (hexane + ethyl acetate) and (cyclohexane + 2-butanone) (table 14) by liquid–liquid extraction. These azeotropes are present in the process of purification of grafted polyolefins [194]. The hexane + ethyl acetate azeotrope was separated by azeotropic batch distillation [195] with heterogeneous entrainers such as methanol, acetonitrile, water, and nitromethane. The other system, cyclohexane + 2-butanone azeotrope, was separated with a dephlegmation membrane [196].

From LLE data for the ternary systems (cyclohexane + 2-butanone + [HMIM][PF₆] or [OMIM][PF₆]), the experimental tie-lines (supporting information, figure S13) show a positive slope where the solute goes preferentially to the solvent-rich phase. Another significant aspect is the fact that the IL does not enter in the organic-rich phase, e.g., the presence of [HMIM][PF₆] and [OMIM][PF₆] was not detected in the upper phase. The selectivity values for the studied ternary systems (supporting information, figure S14) are higher for the [HMIM][PF₆] than for the [OMIM][PF₆],

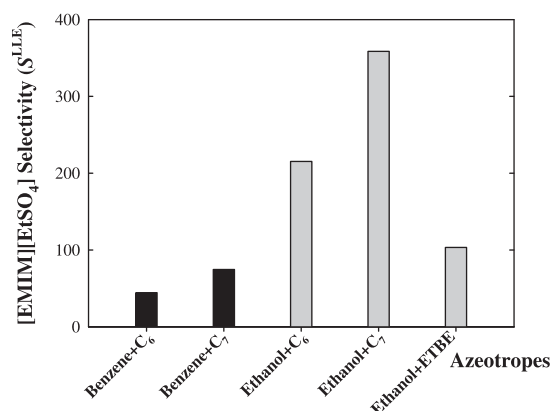


FIGURE 30. [EMIM][EtSO₄] selectivity (S^{LLE}) versus azeotropes separated by liquid–liquid extraction using this IL as solvent [121,157,189] at 298.15 K and solute molar fraction in organic phase ≈ 0.05 .

indicating that the former is a better choice as solvent for these azeotropic separations. Also, the increase of alkyl chain length of the imidazolium decreases the IL's capacity to separate the two azeotropes with hexafluorophosphate.

2.5. Final remarks

Mapping azeotropes as a function of the ILs used as separation agents depicted in figure 1, identified the three most utilized as azeotrope breakers: [EMIM][OTf], [EMIM][EtSO₄], [BMIM][PF₆], [HMIM][PF₆], and [OMIM][PF₆] with nine azeotropes; [BMIM][BF₄] with seven azeotropes; and [BMIM][Cl] with six azeotropes. ILs based on [PF₆][−] and [BF₄][−] anions have received the most research attention. Nevertheless, in the last decade, the use of these ILs has slowed down since they hydrolyse in water and at high temperature [182,183] leading to the formation of HF; their use has consequently been restricted to those systems under water-free conditions and moderate temperatures.

Most research dealing with IL entrainer efficiency in extractive distillation has focused on aqueous azeotropic systems. Five such systems illustrate the effect of ILs on the VLE behavior so that the feasibility of extractive distillation technique can be evaluated: THF + water; ethanol + water; 1-propanol + water; 2-propanol + water; and 2-methyl-2-propanol + water. Figure 29 depicts the influence of the systematic cation and anion variation of imidazolium-based ILs with respect to their entrainer efficiency in extractive distillation for these systems: THF and 2-methyl-2-propanol are the least polar organic solvents and, as can be verified by the selectivities values, they are largely salted-out from the aqueous phase. [EMIM][OTf] and [BMIM][Cl] have received the most vigorous testing as entrainers in extractive distillation. However, in this separation process, the best results are obtained with ILs based in [Cl][−] and [OAc][−] anions. Moreover, the imidazolium salts containing these anions are superior in terms of selective interactions and the increasing relative volatility of the organic solvents. This behavior is in accordance with the fact that the [OAc][−] and [Cl][−] anions show the highest nucleophilicity, e.g., ability to hydrogen bond, of the anions studied. These anions interact with water forming hydrogen bonds, thus decreasing their water activity and, consequently, increasing the relative volatility of the organic solvents.

Ionic liquids based on the alkyl sulfate anion show the greatest potential as solvents in separation by liquid–liquid extraction of the azeotropes reviewed. [EMIM][EtSO₄] is the most frequently studied (figure 1) in this extraction process. Figure 30 compares [EMIM][EtSO₄] selectivities for different azeotropes separated by

liquid–liquid extraction. The best results were obtained for the separation of ethanol with heptane or hexane. From the data presented in figure 20 and tables 10 and 11, it can be concluded that [MMIM][MeSO₄] shows better results than [EMIM][EtSO₄] and [BMIM][MeSO₄].

[MMIM][MeSO₄] and [BMIM][MeSO₄] were used as azeotrope breakers in a laboratory-scale liquid–liquid extraction process and the results also confirmed that [MMIM][MeSO₄] has the highest extraction efficiency [123,124,126,127,129]. These papers confirm the assumption that the IL regeneration and recycling are indeed simple for this process, which is a great advantage relative to traditional extraction solvents.

3. Conclusions and future work

This work reviews the various processes used in the separation of azeotropes. So far, three such processes have implemented ILs as azeotrope breakers: liquid–liquid extraction, extractive distillation and supported liquid membranes. The first two have received the most research attention.

Ionic liquids represent a promising class of neoteric, “green” engineering liquids for separating processes due to their nature as highly selective, nonvolatile azeotrope breakers or azeotrope breakers additives. The IL modular structure allows engineers to tailor properties such as selectivity, viscosity, thermal and chemical stability for specific applications by varying the structure and combination of the constituent ions. Due to the negligible volatility of the ILs, the azeotrope breaker regeneration for extractive distillation and extraction can be carried out by stripping, evaporation, drying, or crystallization. Another advantage of ILs as azeotrope breakers is the reduction of the effects of conventional solvents (VOCs) in global climate change. This approach promises excellent contributions to the reduction of greenhouse gases as targeted in the most recent United Nations Climate Conference (COP15).

This review has analyzed the contribution of different ILs as solvents for the separation of azeotropes by liquid–liquid extraction and extractive distillation. The increase of the alkyl side chain length of the cation/anion plays a negative role in the ILs’ azeotrope breaking ability. This fact has been demonstrated in many cases both in liquid–liquid extraction and extractive distillation and can be taken as a rule of thumb when selecting ILs for the separation of azeotropes. The first step in designing any separation process should take several points into consideration: (i) determination of LLE/VLE data for azeotropic systems + ILs, (ii) evaluation of the separation capacity with decisive parameters such as the distribution coefficients and selectivities for liquid–liquid extraction and the relative volatilities and selectivities for extractive distillation, and (iii) experimental lab-scale extraction/distillation process.

Although only six papers addressed supported liquid membranes for the separation of azeotropes, these data have demonstrated that the use of ILs as membrane solvent enables the supported liquid membrane operation in these processes. The best results are obtained by vapor permeation, which is superior to the reported values for pervaporation.

The systematic analysis of the influence of the structure of the IL on the azeotrope breaking capacity was accomplished in the present review, with the aim of establishing some guidelines for selecting the most suitable IL for the separation of specific azeotropic mixtures. Future work should focus on the influence of the IL structure on the different components of azeotrope systems, in order to develop a systematic method for the task-specific tailoring of ILs for the separation of azeotropic mixtures. A subsequent goal would be the development of a set of industrial requirements for ILs characterizing their industrial suitability as entrainers, extraction solvents, absorption agent, and/or membranes.

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Appendix A. Supplementary data

Supplementary data associated with this article can be found, in the online version, at doi:10.1016/j.jct.2011.05.026.

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